Spanner

Software Requirements Specification v3.0

incorporating the Feasibility Study, Risk Register, System Architecture & Design, and Implementation

ITC327W Assessment 3 - Phase 3: System Development, Implementation and Final Demonstration (Unit 4)
Riese Auto-Clinic - Stakeholder: Michael Riese, Owner/Head Mechanic

GitHub Repository: https://github.com/BlackZ3ro/Spanner-Riese-Auto-Clinic
Live website: https://riese-auto-clinic.vercel.app

Group Y ("The Zealots")

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Version 3.0 - compiled 30 October 2026

Revision History and How to Read This Document

Version	Date	Phase	Summary
1.0	4 Sept 2026	Phase 1 (Units 1-2)	SRS, Feasibility Study and Risk Register as submitted and assessed.
2.0	30 Sept 2026	Phase 2 (Unit 3)	Adds Part B: requirement and scope updates (Changes 7-11), architecture, use case, sequence and entity relationship diagrams, interface designs, design alignment and risk register updates.
2.1	7 Oct 2026	Phase 2	Navigation flow, customer tracking screens, cleaner figures, editable design sources, security finding R-17, Microsoft Project section.
2.2	11 Oct 2026	Phase 2	Team interface designs, client colour choice, updated Microsoft Project schedule, list of open items.
3.0	**30 Oct 2026**	**Phase 3 (Unit 4)**	**Adds Part C: the implementation section. What was built in the Flutter app, the ASP.NET website and the shared Supabase backend; requirement-to-function status for FR-01 to FR-18; two shared processes shown working in both directions with screenshots; access rules; Requirements Change Log Changes 12-18; known limitations; setup guidance; the Phase 3 Microsoft Project schedule; GitHub link.**

As in Phase 2, this is one continuing document. Part A (Phase 1) and Part B (Phase 2) are carried over exactly as in version 2.2, apart from this new cover and revision page. Part C follows them. Where Part C changes an earlier statement it says so and references the earlier section.

Changes in Part C that amend Parts A and B

Earlier reference	Amended by	Effect
Part B, Section 4.2, Change 7 (customer repair-status lookup)	Part C, Change 12	Confirmed by the stakeholder on 30 Oct 2026.
Part B, Section 4.1, FR-05 and FR-06 (labour and cost)	Part C, Change 15	Labour is kept in one place, `booking_labor`, and read by both apps.
Part B, Section 6.3 (Microsoft Project, status at end of Phase 2)	Part C, Section C8	Replaced by the Phase 3 schedule dated 30 Oct 2026.
Part B, Section 6.3 list of open items	Part C, Section C5	Open items are now the known limitations list.

Appendices. Evidence folder: `project-docs/phase-3/evidence/` (26 mobile and 16 website screenshots, captions and an 8-page PDF). Schedule: `project-docs/phase-3/schedule/`. Tracker: `project-docs/phase-3/tracker/`. Unit 4 forms: `project-docs/phase-3/unit-4-forms/`. Code: `mobile/`, `web/`, `supabase/migrations/` (0001 to 0011).

Compiled from group chat content, the group's shared Google Doc, and the Assessment 1 brief. Amber "Assumption/flag" notes mark places where a role or detail was inferred rather than confirmed by the group; please correct these before submission. The Microsoft Project schedule is submitted as a separate file (see Appendices) and is referenced only at a high level here.

Updated per consultation feedback from Ms Mbele (Aug 2026): document justified throughout in Arial with 1.15 line spacing, dedicated cover page added, directly/indirectly affected users clarified (Section 1.2), and the suggestion of customer-facing real-time job tracking evaluated and deliberately excluded from scope (Section 1.10, Change 5; Section 1.11).

1. Software Requirements Specification

1.1 Project Introduction

Spanner (working title) is a workshop management system for Riese Auto-Clinic, a mechanic shop, combining a Flutter mobile application and an ASP.NET web application on a shared Supabase backend. It allows shop staff to manage parts/stock, book in vehicles and assign jobs, capture labour, and generate customer-ready PDF invoices, replacing the shop's current manual, paper- and memory-based process with a single, synchronised system usable from the workshop floor (mobile) or the office (web).

1.2 Stakeholder Profile and Current Process

Organisation: Riese Auto-Clinic, a mechanic shop. Direct stakeholder: Michael Riese, Owner/Head Mechanic, selected for his direct authority over how the shop operates and his ability to give first-hand input into system requirements.

The group's real name and business name are used throughout this document with the stakeholder's knowledge and consent, given the group's direct working relationship with him; this was confirmed as acceptable rather than substituting an anonymised company name.

Directly and Indirectly Affected Users

Following clarification obtained during academic consultation, affected users are classified below by whether they operate the system themselves or are only affected by what it produces:

- Directly affected: Michael Riese (Owner/Head Mechanic), with overall oversight of bookings, stock and invoicing, and may also perform mechanic and front-desk duties personally.
- Directly affected: Mechanics, who log labour time against a job, view assigned jobs and vehicle details, and check parts availability.
- Directly affected: Front-desk/admin staff, who book in vehicles, record parts and stock, generate and hand over invoices, and are typically the ones speaking to a customer during drop-off, consultation and collection.
- Indirectly affected: Shop customers, who do not log into or operate the system themselves, but are affected by what it produces, namely the invoice/quote and any calls or messages staff choose to send them.

Current process: vehicles are booked in verbally, through direct communication with admin staff or the owner. Once a vehicle arrives, mechanics begin diagnostics immediately to identify faults or service points. When parts are needed, a trip is made to a local supplier to source

them, sometimes across multiple stores if one does not stock everything required, before returning to install them and notify the customer of completion or complications. Once work is finished, payment is calculated by combining parts cost, a markup, and labour time, and the job is signed off. The shop keeps customer contact details along with vehicle-identifying information (odometer reading, licence plate and VIN) specifically to protect the business in case a customer becomes non-compliant.

Evidence: informal stakeholder discussion conducted directly with Michael Riese (a personal contact of a group member), documented in the group's shared planning document. A formal, scheduled stakeholder interview (Unit 1 Activity 1.2) confirmed and extended this record; see Section 1.6 and the Requirements Change Log.

1.3 Problem Statement

Current situation: bookings, parts sourcing, labour tracking and invoicing are all handled manually, relying on verbal communication, physical trips to suppliers and the owner's memory or paper notes for cost and stock information.

Problem: manually booking in a vehicle and then sourcing parts from multiple locations, when a single supplier does not carry everything needed, creates unnecessary time delays on jobs. There is no system to track where components were bought for future reference, how much stock of particular parts is held, or labour time spent per vehicle in a structured way.

Consequences: jobs take longer to complete than necessary, restocking decisions are made without good information, and producing an accurate, professional invoice at the end of a job is slower and more error-prone than it needs to be.

Need for improvement: the shop needs a system that documents each job as it comes in, tracks where parts are sourced for future purchasing decisions, maintains a record of stocked parts/service kits, tracks labour time per vehicle, and produces a clear invoice itemising component costs and labour at the end of the job.

1.4 Project Aim and Objectives

Aim: to design and develop an integrated mobile and web-based workshop management system for Riese Auto-Clinic that allows the business to manage parts/stock, book in vehicles and assign jobs, capture labour, and generate customer-ready invoices from a single, synchronised platform.

Objectives:

- Enable staff to record and manage parts/stock inventory, including where a part was bought, its cost price, and the markup/sell price charged to the customer.
- Allow staff to book in a vehicle (capturing customer contact details, VIN, licence plate and odometer reading) and assign one or more jobs to it (e.g. service, engine, suspension).
- Allow staff to capture labour hours worked on a job, together with an hourly labour rate, once the job is marked as finished.
- Automatically calculate the total job cost (parts + labour) when a job is completed.
- Generate a PDF invoice/quote for the completed job that can be sent to the customer via WhatsApp or email, or printed.

- Synchronise data in real time between the Flutter mobile app and the ASP.NET web app using a shared Supabase backend.
- Keep parts sourcing information (supplier, cost) on record to support future purchasing decisions and reduce delays from re-sourcing parts.
- Display a reference image of the vehicle, looked up from brand, model and optional year, so mechanics have a quick visual on what they are working with when a job comes in by phone or message.

1.5 Project Scope (initial)

In scope: a Flutter mobile application and an ASP.NET web application sharing one Supabase backend; parts/stock management (source, cost price, sell price/markup); vehicle booking and job assignment; labour hour and rate capture; automatic job costing; PDF invoice/quote generation shareable via WhatsApp, email or print.

Out of scope: online payment processing; automated customer appointment reminders; support for multiple branches/franchise locations; a customer-facing login portal.

Acceptance criteria: both the mobile and web versions must clearly log the work done on a job and be able to export a PDF invoice for it.

Constraints: the project must be completed within a single semester, with no monetary cost involved in developing it.

Assumptions: the stakeholder (Michael Riese) will remain available for feedback and confirmation throughout the project.

1.6 Stakeholder Needs

The following needs were confirmed directly with Michael Riese through the formal stakeholder interview conducted for Activity 1.2 (transcript below): a combined mobile and web system that keeps track of stock, where parts were sourced and their costs; the same stock visible to all mechanics and to admin, so mechanics know what's available and admin can track money spent per job; a central system linking parts and labour to the specific job and vehicle they relate to, rather than tracking them separately; each mechanic able to log their own labour time against a job, so admin has a clear record of who did what and for how long when preparing an invoice; and an invoice generated automatically once a job is finished, including both labour and parts cost.

Stakeholder Interview Transcript (Activity 1.2)

Conducted 19 August 2026. Present: Gary van Wyk (group) and Michael Riese (stakeholder).

Q1 So you would like us to make your life a bit easier, how can we do that for you?

A1: We would like an app for mobile and a desktop/web version that can keep track of all the stock we keep, where we got the parts, and their costs.

Q2 Okay, that makes sense. And I would assume you would want the app and web version to talk to each other and keep track of the same stock and things?

A2: Yes. We'd like all the mechanics to be able to see what we have available, what needs to get bought, and so on, but also so that the admin personnel can also keep track of the money spent on each job that comes in.

Q3 Alright, so we need to develop a back end database that tracks parts, and for which jobs they're being used for. Might I also make a suggestion? If we are tracking the parts, why not track what types of jobs is being performed on different vehicles? Have a central tracking/management system?

A3: That would be perfect. Having admin and the mechanics all stay in the loop and on the same page with what is being done and the costs involved with it as well.

Q4 That is what I had in mind as well. We can even make it so that every mechanic working on a vehicle is able to log their labour time on the app, admin can see it, and when developing the invoice, have a clear track of who did what on which vehicle and for how long, if that is what would help as well?

A4: That would honestly help us a lot with paperwork and unnecessary admin work.

Q5 Is there anything else we can do for you or that we can include into this system that would improve your situation?

A5: Yes, actually, if it would be possible, to have the system itself export an invoice after all the work is done on a vehicle, which includes all the labour and cost price.

Assumption/flag: The signed stakeholder confirmation letter referenced as evidence for this interview has been collected, countersigned by Michael Riese, and is reproduced in full as Appendix E.

Assumption/flag: The information above (current process, costs and figures) reflects what Michael Riese described directly in discussion and interview; it is not simulated/dummy data. Where specific figures are not yet known (e.g. exact markup percentages, average job values), this will be updated with the shop's actual transaction data as it becomes available, rather than left as a placeholder or generic estimate.

1.7 Functional Requirements

FR-01 The system shall allow staff to register a new vehicle booking with customer contact details, VIN, licence plate and odometer reading.

FR-02 The system shall allow staff to assign one or more job types (e.g. service, engine, suspension) to a booked vehicle.

FR-03 The system shall allow staff to record parts/stock items, including supplier, purchase (cost) price and sell price/markup.

FR-04 The system shall allow staff to link parts used to a specific job on a specific vehicle.

FR-05 The system shall allow staff to capture labour hours worked and an hourly labour rate for a job.

FR-06 The system shall automatically calculate the total job cost (parts + labour) once a job is marked complete.

FR-07 The system shall generate a PDF invoice/quote summarising parts, labour and total cost for a completed job.

FR-08 The system shall allow a generated invoice PDF to be shared via WhatsApp or email, or printed.

FR-09 The system shall synchronise stock, booking and job data between the mobile app and web app in real time via Supabase.

FR-10 The system shall allow authorised staff to view current stock levels of parts/service kits.

FR-11 The system shall maintain a per-vehicle history of past jobs, searchable by VIN or licence plate.

FR-12 The system shall restrict access to shop data to authenticated staff accounts only.

FR-13 The system shall look up and display a reference image of the vehicle from Wikimedia, based on the brand, model and (optionally) year captured at booking, as a best-effort visual aid rather than authoritative data.

1.8 Non-Functional Requirements

NFR-01 (Usability) The interface must be simple enough for mechanics with limited software experience to use with minimal training.

NFR-02 (Performance) Core actions (booking a vehicle, adding a part, generating an invoice) should complete in a few seconds under normal connectivity.

NFR-03 (Availability) The mobile app should tolerate brief connectivity loss without losing unsaved job data, given real connectivity gaps already experienced by the group.

NFR-04 (Security) Customer and vehicle data must be protected via authenticated access and Supabase Row Level Security, and must never be committed to the public GitHub repository.

NFR-05 (Data integrity) Financial calculations (cost, markup, totals) must be accurate and auditable.

NFR-06 (Compatibility) The mobile app must run on common Android/iOS devices used by shop staff.

NFR-07 (Maintainability) The codebase must be organised and documented in GitHub to support future extension beyond this project.

1.9 Integration Requirements

The Flutter mobile app and the ASP.NET web app both connect to a single shared Supabase project (PostgreSQL database, Auth, Storage and Realtime) as the single source of truth for stock, bookings, jobs and invoices. Authentication is handled through Supabase Auth and shared across both clients so staff use one account regardless of which app they open. Changes made on one client (e.g. a new stock entry from the mobile app) propagate to the other (e.g. the web dashboard) via Supabase Realtime subscriptions. PDF invoice generation is triggered on job completion and shared through the device's native share sheet (mobile) or a direct download/share action (web), avoiding the cost and complexity of a dedicated messaging API.

1.10 Requirements Change Log

Change	What Changed	Reason
1	Web application platform specified as ASP.NET rather than a generic web app.	Aligned to the ASP.NET + Supabase technical requirement set out in the Assessment 1 brief.
2	Customer-facing login portal removed from scope.	Operational feasibility: the direct users are shop staff, not customers; economic/schedule feasibility favoured a tighter scope.
3	WhatsApp/email sharing implemented via native device share sheet rather than a paid messaging API.	Economic feasibility: keeps the project at zero development cost and avoids third-party API complexity.
4	Multi-branch/franchise support explicitly excluded.	Riese Auto-Clinic is a single-location business; out of scope was confirmed with the stakeholder relationship in

Change	What Changed	Reason
		mind.
5	Customer-facing real-time job/status tracking explicitly excluded from scope (both mobile and web remain staff-only tools).	Suggested as a possible enhancement during academic consultation, then evaluated directly against stakeholder operational experience (Michael Riese, via Gary). Giving customers a granular status or ETA creates false expectations when parts-sourcing delays are common and outside the shop's control, and would require mechanics to manually log fine-grained progress updates with no operational benefit. All customer-facing output stays limited to the generated invoice/quote and direct calls or messages from staff, matching how the shop already operates.
6	Added a vehicle reference image lookup (FR-13): a Wikimedia-based scraper takes the brand, model and optional year captured at booking and returns a closest-match reference image.	Proposed by Gary as a low-cost visual aid so staff instantly recognise a vehicle from a phone call or message. Uses the free Wikimedia API rather than a paid image service, matching the zero-budget constraint. The scraper is not 100% accurate; better accuracy would require a paid API, which is why the feature is a supplementary visual aid (see R-13) rather than a core requirement.

1.11 Refined Project Scope

Following the stakeholder interview above and the feasibility and risk analysis in Sections 2 and 3, the group finalised the scope below to carry forward into system design.

Project Title: Spanner (RT)

Project Aim: Develop an integrated mobile and web system that allows Riese Auto-Clinic to book vehicles, assign and track jobs, manage parts stock and purchase history, log labour time per mechanic, and generate detailed invoices, all from a shared central system.

Objectives

- Provide mechanics with a simple way to view available stock, log labour time, and record parts used on jobs.
- Allow admin staff to track costs, monitor job progress, and manage stock levels.
- Maintain a single shared source of truth for jobs, parts, labour and costs across both the mobile and web applications.
- Reduce paperwork and administrative effort by automatically generating clear invoices.
- Give staff a quick visual reference of the vehicle via a Wikimedia-based image lookup.

In Scope

- Vehicle booking and job creation.
- Assignment of jobs to mechanics.
- Logging of labour time per mechanic per job.
- Tracking of parts on each job (including supplier and cost).
- Basic stock management (quantity on hand and low-stock awareness).
- Recording of the type of work performed.
- Generation and export of a detailed invoice (PDF) showing parts costs and labour.
- Shared data between the Flutter mobile app and ASP.NET web application via Supabase.

- Vehicle reference image lookup by brand, model and optional year (Wikimedia, best-effort accuracy).

Out of Scope

- Customer-facing portal or client login.
- Customer-facing real-time job/status tracking: raised as a suggestion during academic consultation but deliberately excluded per direct stakeholder operational preference (see Requirements Change Log, Change 5).
- Online payment processing.
- Advanced inventory forecasting or automatic reordering.
- Integration with external accounting software.
- Real-time vehicle tracking.
- Support for multiple workshop branches.

Deliverables

- Flutter mobile application.
- ASP.NET web application.
- Shared Supabase backend and database.
- Ability to export a clear invoice PDF.

Constraints

- Must be completed within a single semester.
- No monetary costs for development tools or services (free tiers only).
- Prototype quality is acceptable (not a fully production-ready system).

Assumptions

- The stakeholder (Michael Riese) and relevant staff will remain available for feedback and meetings throughout the project.
- Mechanics and admin staff will have access to smartphones and computers.
- Basic internet connectivity will be available at the workshop.

2. Feasibility Study

Area	Evaluation
Technical	Flutter (mobile) and Supabase (backend: Postgres, Auth, Storage, Realtime) are already in use, with a working Supabase project and an in-progress mobile prototype giving the group real technical traction. The web client will use ASP.NET, integrating with the same Supabase backend. The main technical gap is the group's limited prior ASP.NET experience, which is a manageable learning curve given the semester timeframe and Supabase handling most backend logic. PDF generation and native share-to-WhatsApp/email are both achievable with free, well-documented libraries. A vehicle reference image lookup (Gary) scrapes Wikimedia for a closest-match image based on brand, model and optional year, giving mechanics a quick visual reference; it is a supplementary feature layered on top of the core requirements, not a dependency for them. Conclusion: technically feasible, with ASP.NET familiarisation flagged as a schedule risk (see Risk Register).
Operational	The proposed workflow (book a vehicle, assign a job, capture parts and labour, generate an invoice) maps directly onto Riese Auto-Clinic's actual process as described by Michael Riese, so no retraining in a new way of working is required; only digitising steps staff already perform. The main operational requirement is a simple, low-friction interface, since the direct users are

Area	Evaluation
	mechanics and admin staff rather than technical staff. Conclusion: operationally feasible provided the interface stays simple (see NFR-01).
Economic	The project uses free-tier tooling throughout: Flutter and ASP.NET are free and open-source, Supabase's free tier covers a project of this scale, GitHub is free for the group's use, and the vehicle reference image lookup uses the free Wikimedia API rather than a paid image service. Sharing invoices uses the device's native share sheet rather than a paid WhatsApp Business API or SMS gateway. This matches the stated project constraint of zero development budget. Conclusion: economically feasible at no direct cost to the group or the stakeholder.
Schedule	The group has a head start (mobile prototype and Supabase project already created before the module formally began) and a single-semester deadline. The realistic risk is the web application (ASP.NET) and full mobile-web-Supabase integration, which have not yet started. Conclusion: feasible only if the remaining scope is kept tight, as the refined scope below deliberately excludes lower-priority features to protect the schedule.
Legal	The system stores personal customer contact details and vehicle-identifying data (VIN, licence plate, odometer) that Riese Auto-Clinic already collects for its own compliance protection. This data must be handled in line with South Africa's Protection of Personal Information Act (POPIA): collected only for the stated business purpose, access-restricted to authenticated staff via Supabase Row Level Security, and never committed to the public GitHub repository. Conclusion: legally feasible provided access control and data-minimisation practices in NFR-04 are implemented.
Ethical	The system must represent parts costs and labour charges to customers accurately and transparently, since the invoice is the customer-facing record of what they are charged. Stakeholder and customer information must only be used for the stated project and assessment purpose. Conclusion: ethically feasible with no material concerns identified, provided invoice calculations remain accurate and auditable (see NFR-05).

Overall Feasibility Decision

The project is feasible with changes. It is well supported technically and economically by the group's existing Supabase project and mobile prototype, and it is operationally and ethically sound because it digitises a process the stakeholder already follows. The two conditions attached to this decision are: (1) the ASP.NET web application must start in parallel with continued mobile work rather than after it, given the group's limited prior ASP.NET experience (see R-03), and (2) the scope must stay to the refined list in Section 1.11 to protect the single-semester schedule.

3. Project Risk Register

Likelihood and Impact are rated Low / Medium / High (1–3); Score = Likelihood × Impact, bucketed as Low (1–2), Medium (3–4) or High (6–9).

ID	Risk Statement	Category	Likelihood	Impact	Score / Level	Mitigation	Contingency	Owner	Status
R-01	Michael Riese's availability for feedback and confirmation reduces as the semester progresses, delaying requirement sign-off.	Stakeholder	Medium	High	6 / High	Agree a light-touch, low-effort check-in channel (WhatsApp) early, and batch questions to minimise his time burden.	Fall back on previously confirmed requirements and document any assumption explicitly for later confirmation.	Gary	Open
R-02	The stakeholder requests additional features once he sees the working prototype, expanding scope beyond what the semester allows.	Scope	Medium	Medium	4 / Medium	Maintain a clearly refined, agreed scope (below) and log any new request in the requirements change log rather than building it immediately.	Park new requests as a 'Phase 2 / future work' list to revisit after submission.	Denny	Open
R-03	The web application (ASP.NET) has not yet started and the group has limited prior ASP.NET experience, risking a late or rushed build. Ownership sitting with Gary alongside mobile and Supabase also concentrates technical work on one person.	Technical	Medium	High	6 / High	Start ASP.NET scaffolding and the Supabase connection early in Phase 2, in parallel with mobile work rather than after it; consider pulling in Khumo or Kea for support if Gary's workload becomes a bottleneck.	Reduce the web app to a read-only dashboard of the same Supabase data if full parity can't be reached in time.	Gary	Open
R-04	The project outgrows Supabase's free-tier storage/bandwidth limits before the project is demonstrated.	Technical / Infrastructure	Low	Medium	2 / Low	Monitor usage via the Supabase dashboard monthly; avoid storing large files (e.g., full-resolution photos) unnecessarily.	Compress or offload large assets, or apply for Supabase's student/hobby tier if a limit is approached.	Gary	Open
R-05	Real-time sync between the mobile app and web app produces conflicting or duplicate records (e.g., two staff editing the same job).	Technical / Integration	Low	Medium	2 / Low	Use Supabase's row-level timestamps and design simple, single-writer-per-record flows where possible.	Add a manual 'last updated by' field so conflicts are visible and can be corrected by staff.	Gary	Open
R-06	A group member loses connectivity or availability at short notice (already	Team / Resource	Medium	Medium	4 / Medium	Keep the progress tracker current so gaps are visible early, and	Reassign or share the affected task temporarily among members with	Wisani	Monitoring

ID	Risk Statement	Category	Likelihood	Impact	Score / Level	Mitigation	Contingency	Owner	Status
	experienced once this phase), delaying their task.					maintain a WhatsApp-first coordination habit that doesn't depend on any one channel.	capacity.		
R-07	Task completion becomes uneven across members as deadlines approach, with some falling behind.	Team / Schedule	Medium	Medium	4 / Medium	Review the Task Progress tracker weekly and flag slipping tasks before they become blockers.	Redistribute the task or bring in support from a member with lighter load (as already planned for Khumo supporting Gary).	Wisani	Monitoring
R-08	The formal stakeholder interview (U1-05) and supporting evidence are not finalised in time to support the SRS.	Documentation / Evidence	Low	High	3 / Medium	Schedule the interview as a fixed WhatsApp call this week rather than leaving it open-ended, given direct access to the stakeholder.	Use the existing informal stakeholder discussion as interim evidence, clearly logged as informal pending the formal record.	Gary	Open
R-09	The GitHub repository is not set up in time, leaving no evidence of individual contribution as required by the assessment.	Project Management	Low	High	3 / Medium	Kea has taken ownership of repository setup and should create it with a proper README promptly.	If delayed, migrate existing local project history into the repository as soon as it is created so contribution history is not lost.	Kea	Open
R-10	Customer and vehicle data (contact details, VIN, licence plate) is exposed through inadequate access control or accidentally committed to GitHub.	Legal / Compliance	Low	High	3 / Medium	Apply Supabase Row Level Security from the start and add real stakeholder data to .gitignore / exclude from any repository content.	Rotate any exposed credentials immediately and review repository history to remove the exposure if it occurs.	Denny	Open
R-11	PDF invoice generation does not integrate cleanly with the chosen library, producing incorrectly formatted or incomplete invoices.	Technical	Low	Medium	2 / Low	Prototype the PDF generation step early with sample data rather than leaving it until the end of development.	Fall back to a simpler, text-based PDF layout if a more polished template proves too time-consuming to implement.	Gary	Open
R-12	The Microsoft Project schedule underestimates remaining effort, causing later phases to slip.	Schedule	Medium	Medium	4 / Medium	Base task durations on actual Phase 1 pace rather than optimistic estimates, and review the schedule against real progress each	Cut lower-priority scope items (see refined scope) rather than compressing testing or integration time.	Denny	Open

ID	Risk Statement	Category	Likelihood	Impact	Score / Level	Mitigation	Contingency	Owner	Status
						phase.			
R-13	The Wikimedia-based vehicle image lookup returns an inaccurate or missing reference image, since the free scraper is not 100% accurate by design.	Technical	Medium	Low	2 / Low	Treat the image as a best-effort visual aid only, never authoritative vehicle data; fall back to showing the brand/model/year as plain text if no match is found.	If accuracy proves too poor to be useful, disable the auto-image feature and keep vehicle identification text-only, or evaluate a paid image API in a future phase if budget allows.	Gary	Monitoring

Three Highest Project Risks

- R-01: Stakeholder availability declining over the semester (Score 6/High). Everything downstream depends on continued access to Michael Riese.
- R-03: ASP.NET web app not yet started, limited prior experience (Score 6/High). This is the single biggest technical gap between current progress and the full scope.
- R-12: Microsoft Project schedule underestimates remaining effort (Score 4/Medium). This is the next-highest risk, and directly relevant once the current draft schedule is reviewed against actual Phase 1 pace.

Assumption/flag: R-03 (ASP.NET web app) is confirmed to Gary, who owns the mobile app and Supabase setup and will also build the web app. R-09 (GitHub setup) is confirmed to Kea, who volunteered for it directly. R-13 (vehicle image lookup accuracy) is a new, low-severity risk added alongside the feature itself; it is being monitored rather than treated as a blocker.

Appendices

Appendix A: GitHub Repository. https://github.com/BlackZ3ro/Spanner-Riese-Auto-Clinic. Contains the group's source code, folder structure and commit history as evidence of individual contribution.

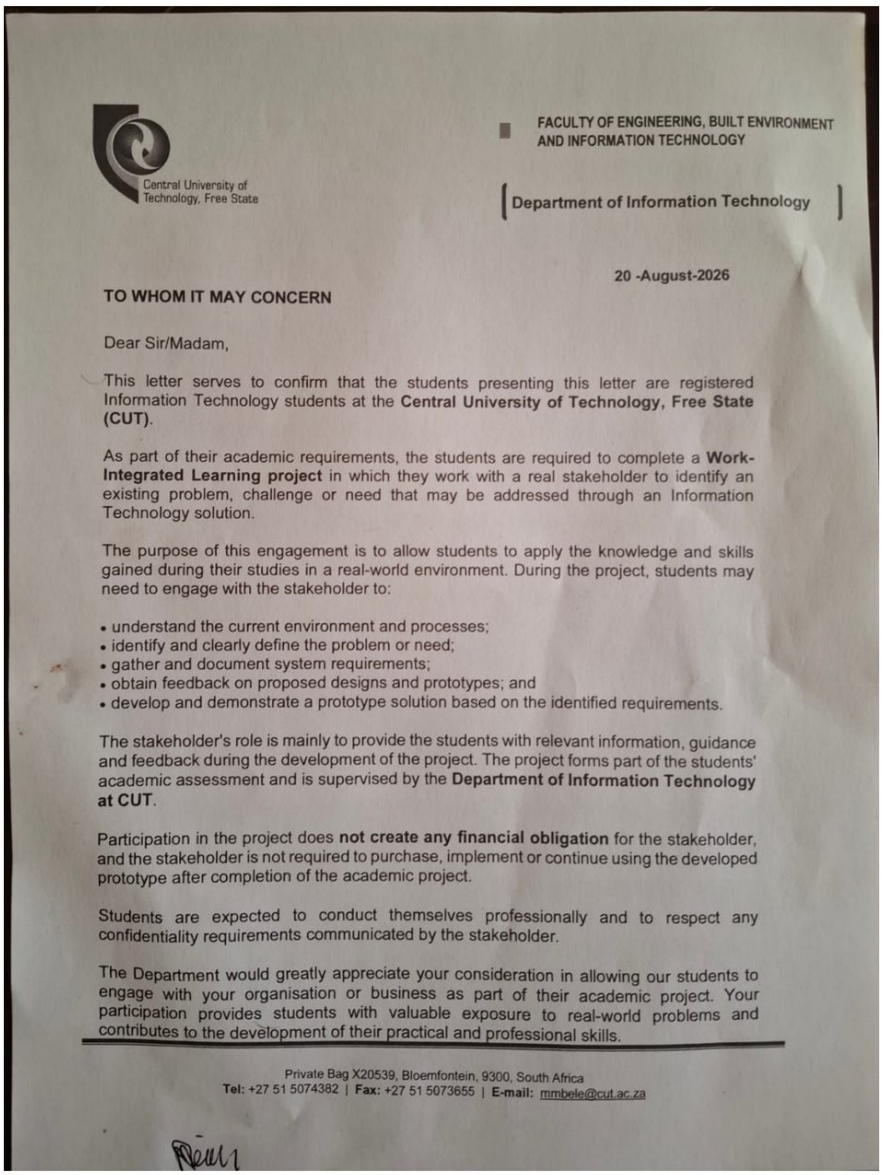
Appendix B: Stakeholder Interview Transcript. Included in full in Section 1.6 above (Activity 1.2, conducted 19 August 2026 with Gary van Wyk and Michael Riese).

Appendix C: Academic Consultation Notes. Feedback received from Ms Mbele (module mentor) on this document was reviewed and applied throughout; see the formatting and content notes referenced in Sections 1.2, 1.6, 1.10 and 1.11.

Appendix D: Microsoft Project Schedule. Submitted as a separate file (TheZealotsSchedule.mpp) alongside this document and the Progress Checker, per the submission instructions; not reproduced here as it is not a static document.

Appendix E: Signed Stakeholder and Departmental Confirmation Letter. Reproduced in full overleaf. Confirms the group's Work-Integrated Learning engagement with the Department of Information Technology at CUT and with the stakeholder, Michael Riese of Riese Auto-Clinic, and carries both the department's digital signature (Mpho Mbele, Departmental Manager, 20 August 2026) and Michael Riese's handwritten signature.

Appendix E: Signed Stakeholder and Departmental Confirmation Letter



Central University of Technology, Free State

FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

Department of Information Technology

20 -August-2026

TO WHOM IT MAY CONCERN

Dear Sir/Madam,

This letter serves to confirm that the students presenting this letter are registered Information Technology students at the **Central University of Technology, Free State (CUT)**.

As part of their academic requirements, the students are required to complete a **Work-Integrated Learning project** in which they work with a real stakeholder to identify an existing problem, challenge or need that may be addressed through an Information Technology solution.

The purpose of this engagement is to allow students to apply the knowledge and skills gained during their studies in a real-world environment. During the project, students may need to engage with the stakeholder to:

- understand the current environment and processes;
- identify and clearly define the problem or need;
- gather and document system requirements;
- obtain feedback on proposed designs and prototypes; and
- develop and demonstrate a prototype solution based on the identified requirements.

The stakeholder's role is mainly to provide the students with relevant information, guidance and feedback during the development of the project. The project forms part of the students' academic assessment and is supervised by the **Department of Information Technology at CUT**.

Participation in the project does **not create any financial obligation** for the stakeholder, and the stakeholder is not required to purchase, implement or continue using the developed prototype after completion of the academic project.

Students are expected to conduct themselves professionally and to respect any confidentiality requirements communicated by the stakeholder.

The Department would greatly appreciate your consideration in allowing our students to engage with your organisation or business as part of their academic project. Your participation provides students with valuable exposure to real-world problems and contributes to the development of their practical and professional skills.

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Central University of Technology, Free State

FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

Department of Information Technology

Should you require confirmation or further information regarding this project, please contact me directly using the details below.

Digitally signed by Mpho Mbele
Date: 2026.08.20 13:29:10 +02'00'

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Part B - Phase 2: System Architecture and Design

Part B records how the requirements in Part A were turned into a design, and what changed on the way. The design was drawn from the system as actually built (the Flutter app, the ASP.NET app and the shared Supabase database in the GitHub repository), not invented separately, so every diagram below can be checked against running code.

4. Requirement and Scope Updates

4.1 New Functional Requirements (continues Section 1.7)

ID	Requirement	Source
FR-14	The system shall let a customer check their vehicle's repair status by entering the booking's reference code, without logging in. The customer sees the vehicle, its current phase, and the parts and labour lines on the job, but no staff data and no other booking. Available in the mobile app ("I'm a customer") and on a public web page (/Track).	Academic consultation, 22 Sept 2026 (Change 7)
FR-15	The system shall let staff move a booking through five phases: Book-in & Inspection, Labor in progress, Reassembly, Finished - ready for pickup, Finalized - booked out to customer. Both apps offer the phase change. "Finalized" is set automatically when a booking is closed.	Change 8
FR-16	The system shall send a push notification to a customer's phone when their booking's phase changes, if the customer opted in on that device.	Change 8
FR-17	Only an Admin shall be able to finalize a booking, delete a booking, reopen a closed booking, or remove a part or labour line. Mechanics can do everything else.	Group decision, 23 Sept 2026 (Change 9)
FR-18	Deleting a booking shall return every allocated part to stock and remove its labour in a single all-or-nothing operation.	Design, supports NFR-05 (Change 11)

FR-12 (authenticated staff access) is now met on both clients: mobile gained Supabase login on 28 Sept 2026 (Change 10).

4.2 Requirements Change Log - continued (continues Section 1.10)

#	What changed	Reason
7	**Customer repair-status lookup added (FR-14). Partially reverses Change 5**, which excluded customer-facing real-time job/status tracking.	Raised again and confirmed at the Phase 2 academic consultation with Ms Mbele (22 Sept 2026): customers "don't need to login, even just a screen with updates". The design answers Change 5's two original objections directly. (1) **False expectations:** the customer sees only five coarse phases, never an ETA or a promised completion date, so parts-sourcing delays do not create broken promises. (2) **Extra work for mechanics:** phases are changed at natural milestones with one tap, not as fine-grained progress logging. The Phase 1 exclusion of a customer *login portal* (Change 2) still stands: the lookup uses a one-off random reference code, not an account. **Stakeholder confirmation:** Change 5 was made on Michael Riese's operational preference, so the group took this reversal back to him. **Michael Riese confirmed Change 7** (confirmation reported to the group on 11 October 2026).
8	Booking phases (FR-15) and phase-change push notifications (FR-16) added.	Makes the FR-14 lookup useful without the customer repeatedly checking. Uses Firebase Cloud Messaging's free tier and a Supabase Edge Function, so the zero-budget constraint (Section 2, Economic) still holds.
9	Admin and Mechanic roles with admin-only actions (FR-17).	Finalizing, deleting and reopening a booking change money and stock records; Part A's objective that "admin staff track costs" (1.11) needs those actions controlled. Agreed by the group on 23 Sept 2026 and enforced in both apps, with deletes also refused by the database for non-admins.

10	Mobile app changed from a role picker to real Supabase Auth login, with the role read from a `profiles` table.	Brings the mobile app in line with FR-12 and NFR-04; the earlier picker let anyone choose "Admin".
11	Reopen a closed booking (admin only, re-enter password) and atomic delete-with-restock (FR-18) added on the web app.	Mistakes happen after a job is finalized; reopening needs a deliberate, authenticated step. Doing restock and delete in one database transaction keeps stock accurate (NFR-05).

4.3 Refined Scope - amendment (continues Section 1.11)

- **Moved into scope:** a login-free repair-status lookup by reference code showing coarse phases, with optional push notifications (Changes 7 and 8).
- **Still out of scope:** a customer portal or customer accounts; live or granular progress tracking; ETAs or promised completion times; online payment; multi-branch support; accounting integration.
- **Deliverables** are unchanged: Flutter app, ASP.NET app, shared Supabase backend, exportable invoice PDF.

4.4 Risk Register - updates (continues Section 3)

ID	Update	Owner	Status
R-03	ASP.NET web app risk: the app was built in milestones M0-M8 (login, dashboard, bookings, stock, labour costing, invoice PDF, vehicle history, realtime, testing).	Gary / Wisani	Closed
R-02	Scope growth **materialised**: the consultation added FR-14 and the build added FR-15/16. Handled as R-02 intended, through the change log (Changes 7-8) rather than silently.	Denny	Managed
R-10	**Materialised 28 Sept 2026:** a Firebase service-account key was committed to the GitHub repository. The file was removed and the pattern added to `.gitignore` the same day; because it remains in git history, the key must be revoked and replaced in Google Cloud, exactly as R-10's contingency requires.	Gary	Open - revocation to be confirmed
R-14	**New.** Michael Riese may not accept Change 7, since Change 5 was based on his preference. Mitigation: reconfirm before submission, showing the coarse-phase, no-ETA design. Contingency: disable notifications and keep the lookup staff-facing only.	Denny	Closed - Michael Riese confirmed Change 7 (11 Oct 2026)
R-15	**New.** Two sources of truth for "who is an admin": the web app's configured admin list and the database `profiles` table. If they drift, a user could be admin in one app and not the other. Mitigation: keep one admin account list and check both when staff change.	Wisani	Open
R-17	**New, found 7 Oct 2026.** The staff function that changes a booking's phase could be called **without logging in** using the public app key, because Postgres lets everyone execute new functions unless that is revoked. Someone with a booking's internal id could change its phase (sending the customer a notification) and read the booking row. The delete function was already protected. Fix: migration 0008 revokes anonymous access to all staff functions; the customer lookup stays public by design.	Gary	Open - 0008 to be run
R-16	**New.** Labour is recorded in two tables: mobile uses `booking_labor`, web uses `jobs`. Invoice totals can differ depending on which app produced them. Mitigation: agree one labour table before Phase 3 and migrate the other.	Gary	Open

5. System Design

5.1 System Architecture (Activity 3.1)

Spanner is one integrated system with two staff clients and one shared backend. Both the Flutter app and the ASP.NET app talk to Supabase directly; there is no separate custom API layer. Supabase provides authentication, the Postgres database protected by Row Level Security, the REST (PostgREST) interface, database functions for multi-step operations, and Realtime updates to the web dashboard. The only server-side code outside the two apps is one Supabase Edge Function that sends phase-change notifications through Firebase Cloud Messaging.

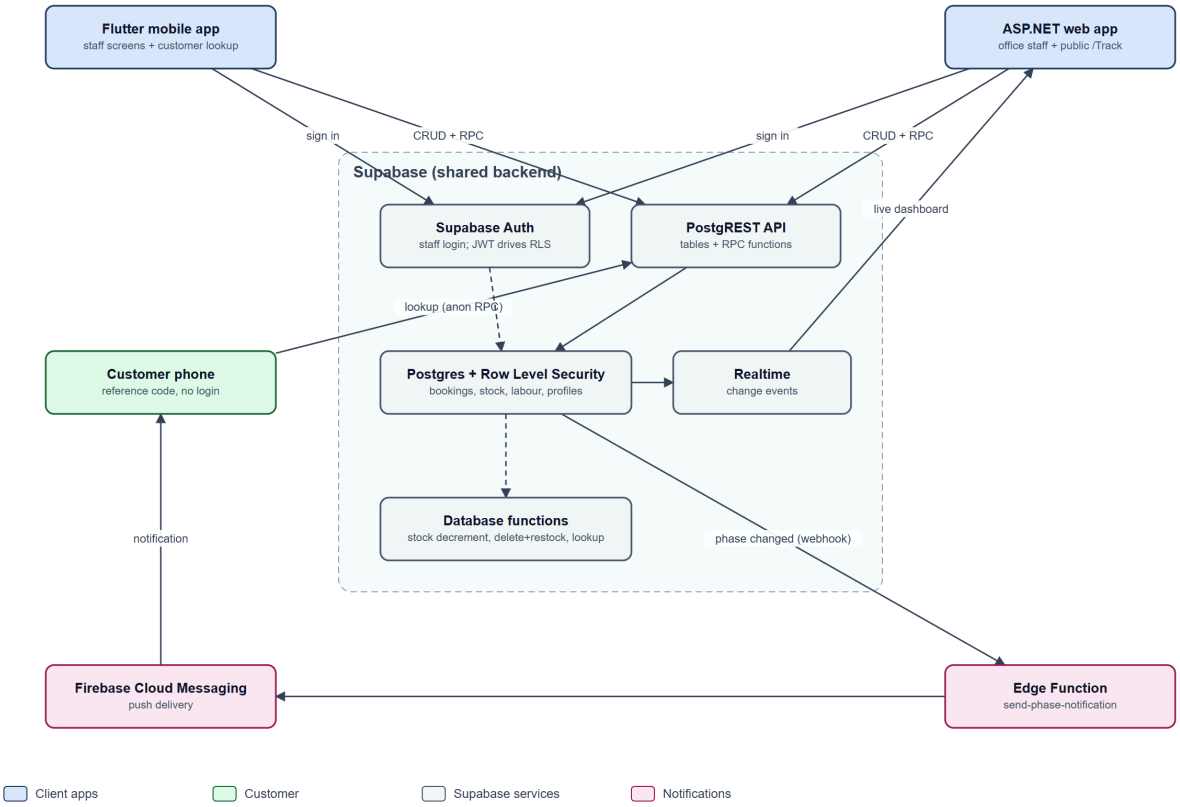


Figure 1 - System architecture (editable source: architecture.drawio)

Component	Role
Flutter mobile app	Workshop-floor client for staff (bookings, phases, parts, labour, stock, invoice export) and the login-free customer lookup screen.
ASP.NET web app	Office client for staff: dashboard with live updates, bookings, stock, invoices, vehicle history, admin-only actions.
Supabase	Single source of truth: Auth, Postgres + Row Level Security, PostgREST, database functions (e.g. atomic stock decrement, delete-with-restock, reference-code lookup), Realtime.
Edge Function + Firebase Cloud Messaging	Triggered by a database webhook when a booking's phase changes; sends a push notification to subscribed customer devices.

Key data flows. (1) A booking created on either app is immediately visible on the other, because both read and write the same tables. (2) Adding a part decrements stock inside one database function, so a failure cannot lose stock. (3) Admin-only actions are hidden in both apps and rejected server-side; deletes are also refused by the database for non-admins. (4) The customer lookup is the only unauthenticated path: it calls a narrowly scoped database function, and anonymous users have no direct read access to any table.

5.2 Use Case Diagram (Activity 3.2)

The use case diagram follows the structure agreed at the 22 Sept consultation: Admin on the left and Mechanic on the right, each shared use case connected to both, and admin-only use cases connected to Admin alone. Login includes validation with Supabase Auth and a dashboard, which is a landing page, not a hub other use cases return to. The Customer is a separate actor with no login.

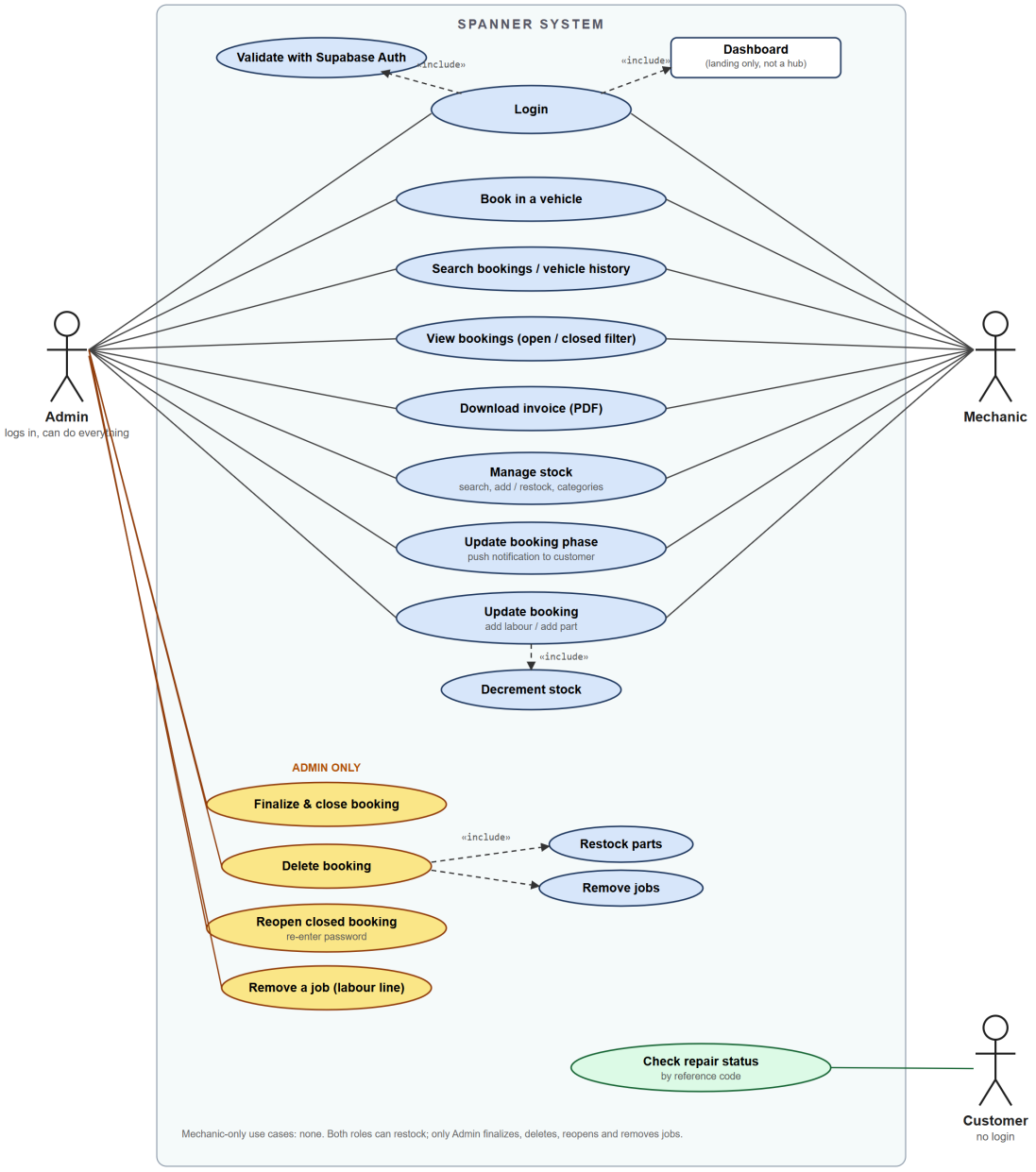


Figure 2 - Use case diagram

Use case	Mechanic	Admin
Login; book in a vehicle; search bookings / vehicle history; view bookings (open/closed filter); download invoice; manage stock (incl. restock); update booking (add labour/parts); update booking phase	Yes	Yes
Finalize & close booking; delete booking (includes restock parts, remove jobs); reopen closed booking (re-enter password); remove a part or labour line	No	Yes
Check repair status by reference code (Customer, no login)	Customer only	

5.3 Sequence Diagrams (Activity 3.2)

Every sequence diagram starts with authentication and then shows the work, and uses **alt** frames (one branch or the other) for decisions. Five diagrams are included: at least one for each client, as the brief requires.

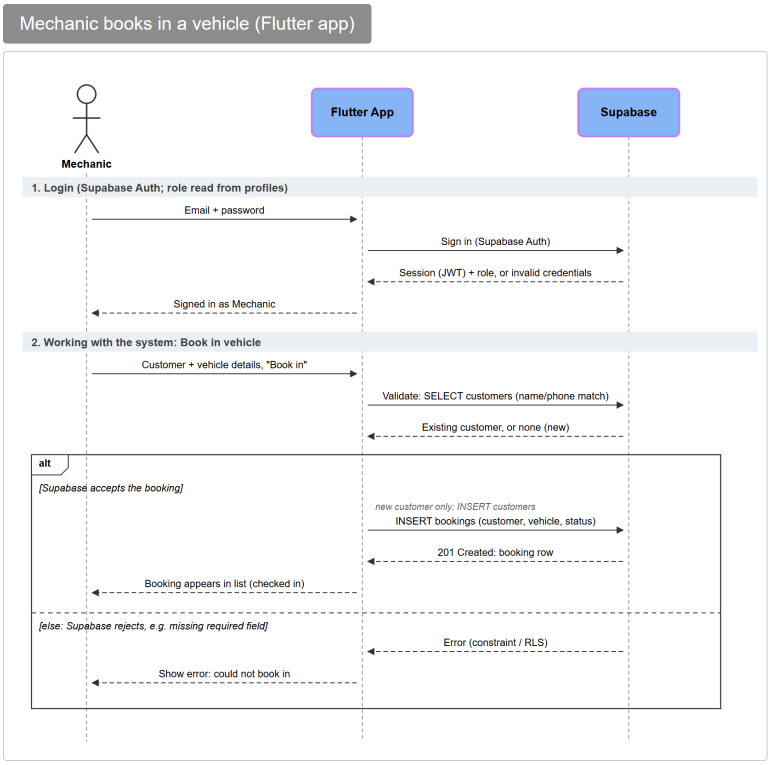


Figure 3 - Mechanic books in a vehicle (Flutter app)

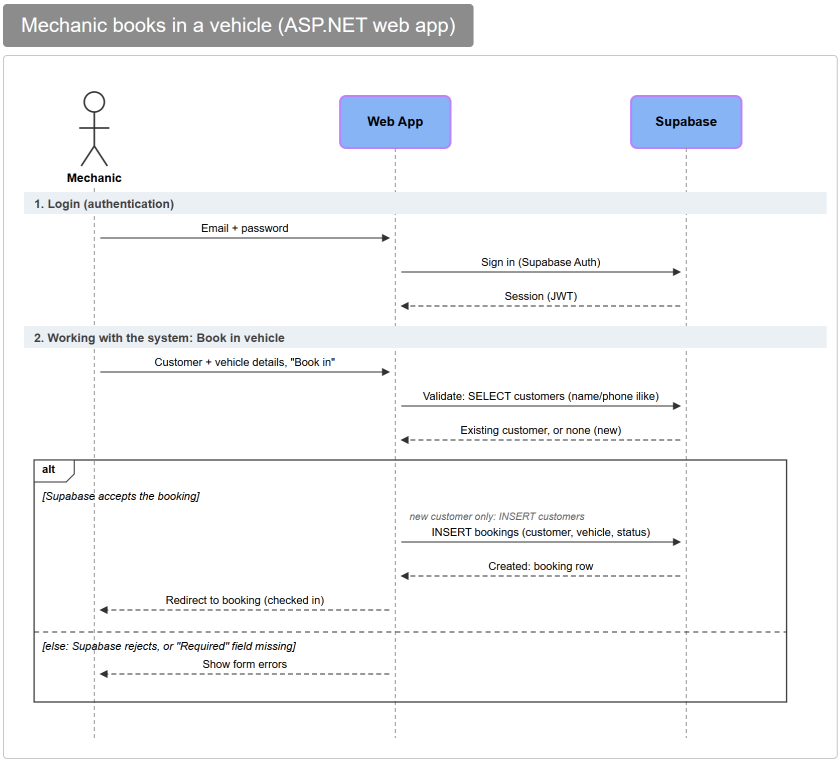


Figure 4 - Mechanic books in a vehicle (ASP.NET web app)

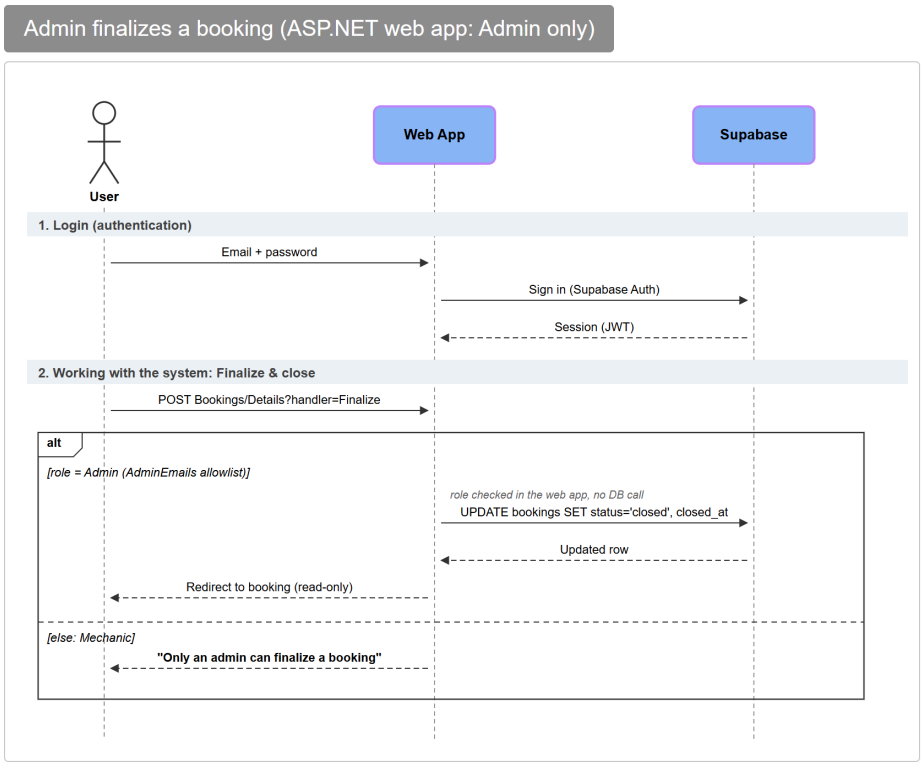


Figure 5 - Admin finalizes a booking (the admin check happens in the web app, not as a database call)

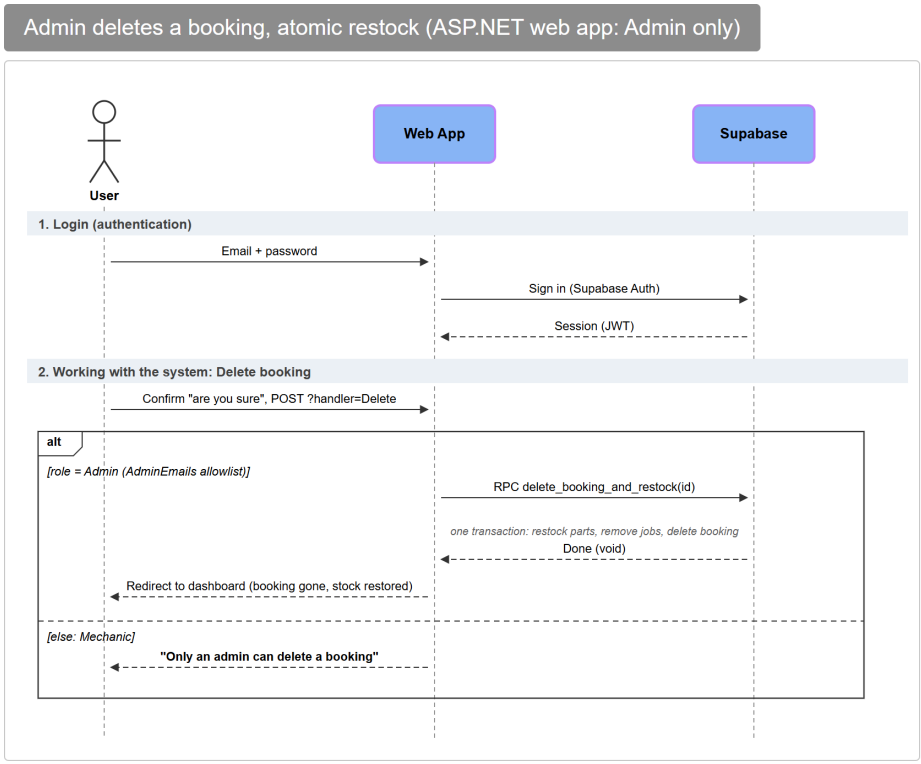


Figure 6 - Admin deletes a booking with atomic restock

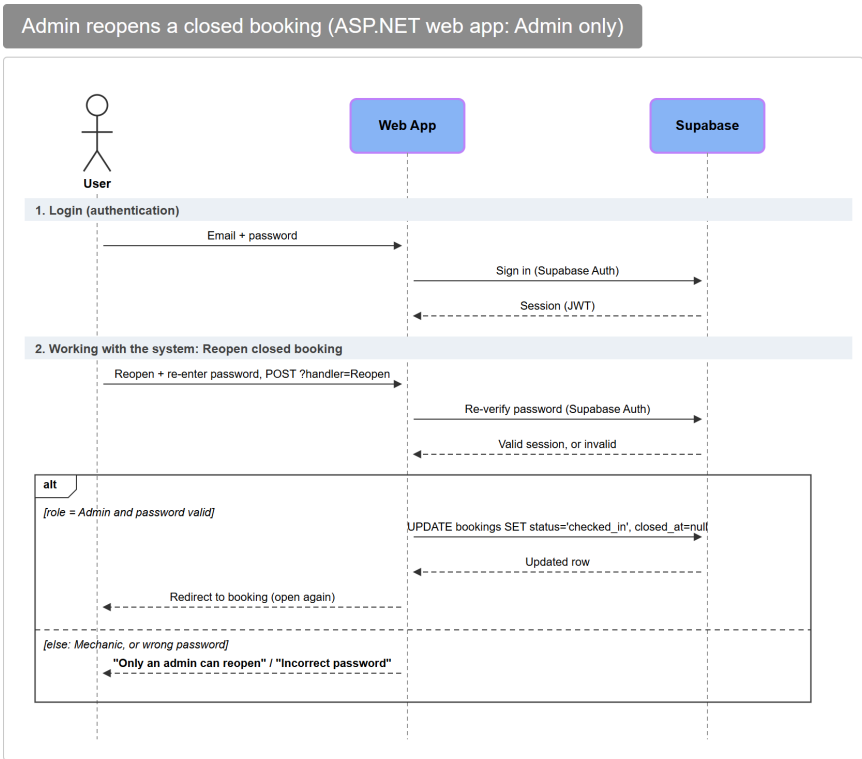


Figure 7 - Admin reopens a closed booking (password re-verified)

5.4 Entity Relationship Diagram and Database Design (Activity 3.2)

The ERD reflects the live Supabase schema. `bookings` is the centre of the model: it links a customer, the parts used (`booking_parts`, which reference `stock_items`), labour (`jobs` on web, `booking_labor` on mobile, see R-16), and customer push subscriptions. Every stock change is also written to `stock_movements` as an audit trail (NFR-05). Staff roles live in `profiles`, one row per authenticated user.

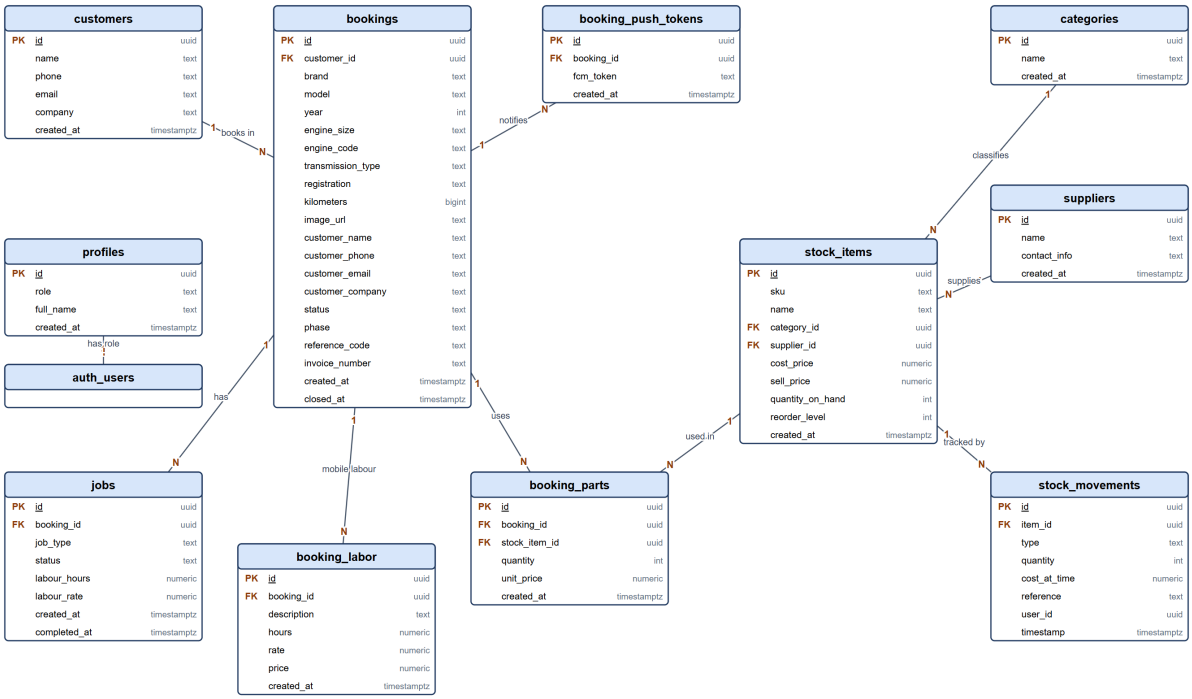


Figure 8 - Entity relationship diagram: primary and foreign keys, and 1/N cardinality on every relationship (editable source: erd.drawio)

Rule	How the database enforces it
Stock can never go negative through a part allocation	Single database function decrements and records the movement in one transaction.
Deleting a booking restores stock exactly	One function restocks, logs, removes labour and parts, then the booking (FR-18); non-admins refused.
Customers see only their own booking	Random, unique `reference_code`; lookup through a security-definer function returning a fixed, minimal shape.
Only staff read or change shop data	Row Level Security on every table (NFR-04).

5.5 Flutter Interface Design (Activity 3.3)

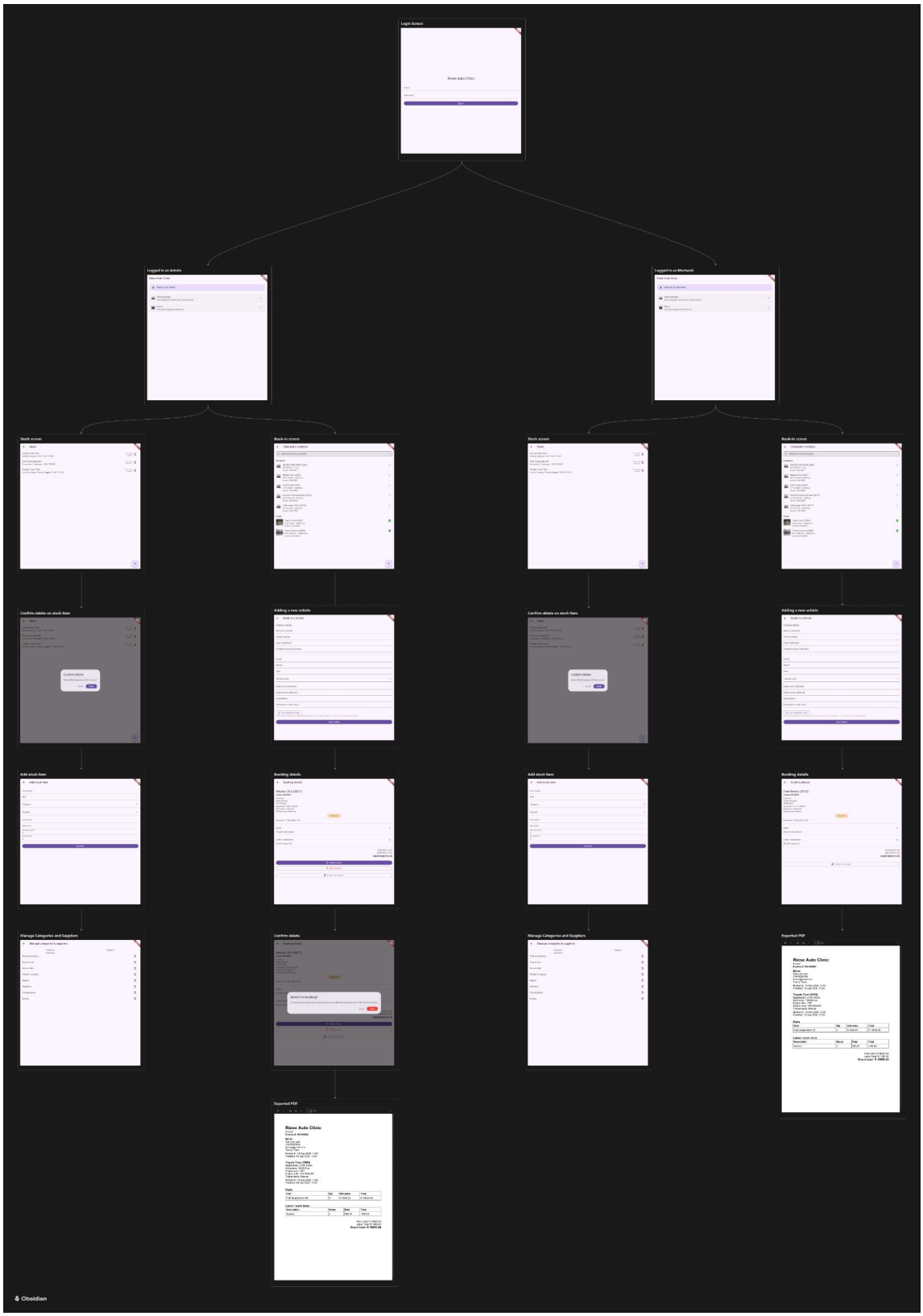


Figure 9 - Flutter wireframe flow by Gary van Wyk (login, admin and mechanic branches, stock, book-in, booking details, confirmations, exported invoice). Drawn before the customer screens were added on 28 Sept.

5.5.0 Flutter screen mockups by Gary van Wyk (11-13 Oct)

These mockups are Gary's own work. He turned his wireframe flow into high-fidelity mockups of the five main screens in **Obsidian Canvas**, using a dark theme in the client's colours: **gold** for the brand and main actions, **green** for done, and **orange** for the active phase or anything that needs attention. Each screen has a note naming the Flutter file it belongs to (for example `booking_detail_screen.dart`), and arrows show the navigation: the role choice leads a customer to the vehicle status and staff to home, then the bookings list, then booking details. Gary's notes also say honestly which parts are already built and which are design ideas not built yet (the open-jobs and low-stock strip on home, the coloured phase tags, and the step tracker on booking details).

The booking detail screen shows every value a mechanic captures: customer, vehicle, phase, parts and labour lines with running totals, and the admin-only "Finalize & close". Each value maps to a column in the ERD (Fig. 8).

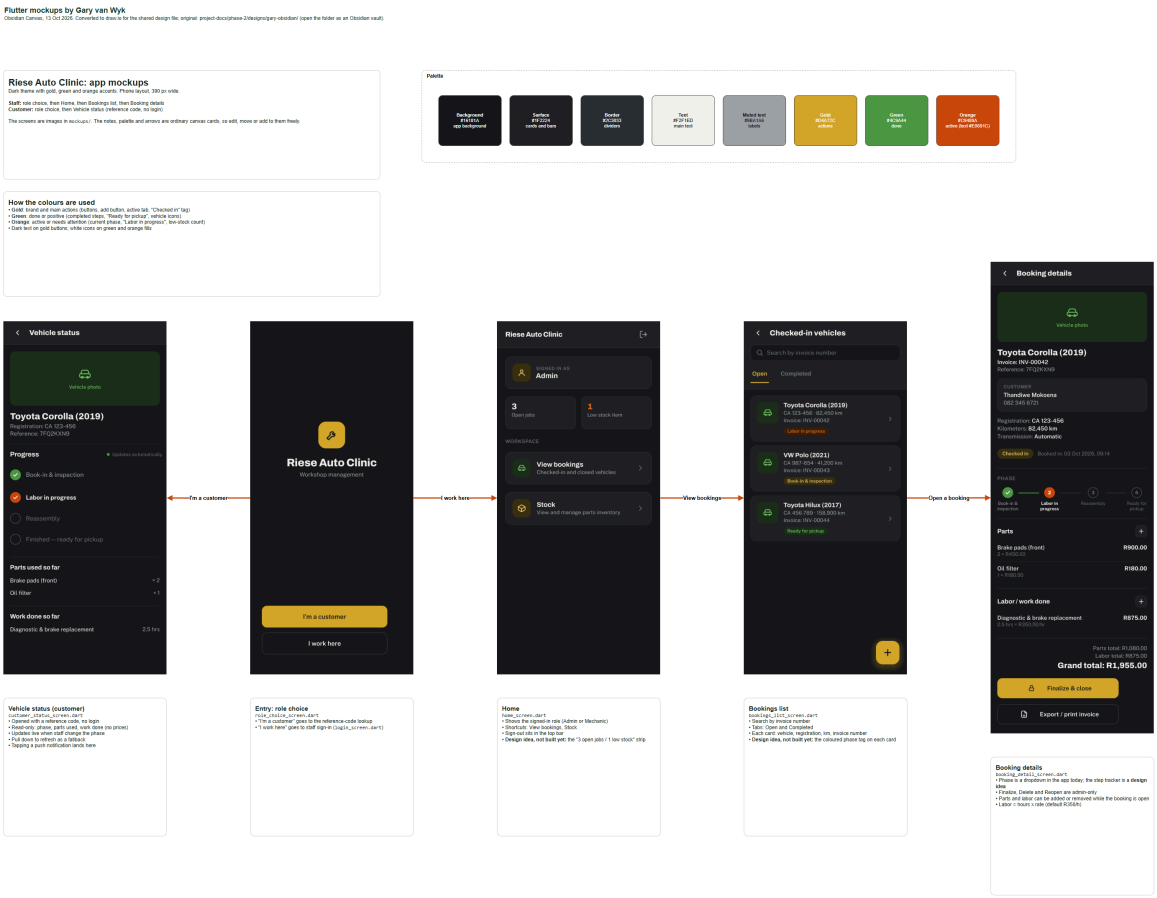


Figure 9a - Flutter mockups by Gary van Wyk (Obsidian Canvas, 13 Oct 2026): vehicle status (customer), role choice, home, bookings list, booking details, with his notes, colour palette and navigation. Editable source: Riese Auto Clinic Mockups.canvas (GitHub: project-docs/phase-2/designs/gary-obsidian); also page 12 of spanner-all-designs.drawio. Sample data.

So that every interface design has an editable source in a standard design tool, the mobile screens were redrawn in draw.io in the client's colours, with the navigation between them (Fig. 9b). It is page 10 of `spanner-all-designs.drawio`.

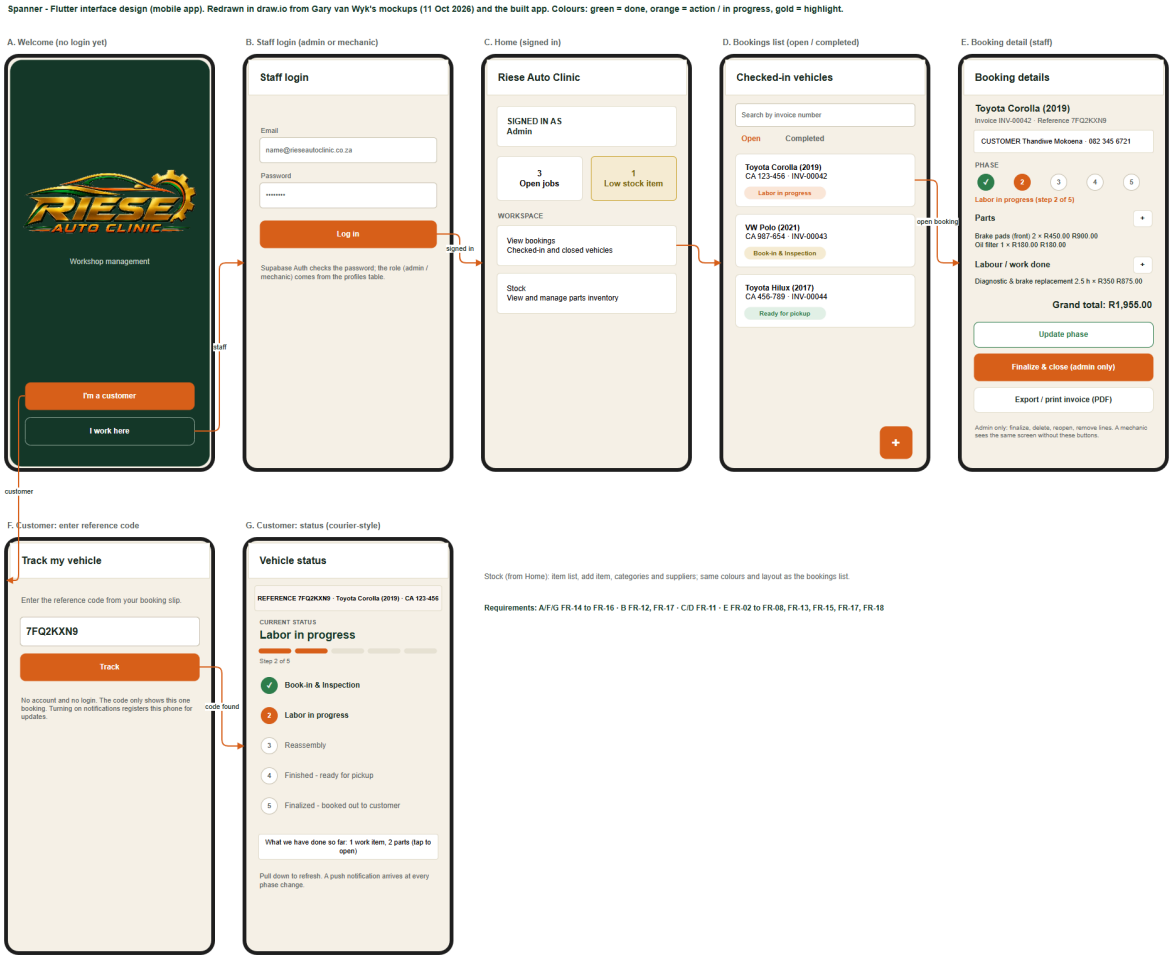


Figure 9b - Flutter interface design in draw.io: A welcome, B staff login, C home, D bookings list, E booking detail, F customer code entry, G customer status (editable source: flutter-screens.drawio).

5.5.1 Customer tracking screen (mobile)

The customer screen is designed like a courier tracking page (for example PUDO or The Courier Guy), because customers already understand that pattern: a strip with their reference code and vehicle, one large "current status", a five-step progress bar and a timeline of the stages. The customer reaches it from the welcome screen ("I'm a customer"), enters the reference code from their booking slip, and can pull down to refresh; push notifications announce each change. The screenshots are the real Flutter screen rendered from the project code with sample data. It compiles with no analyzer errors and is with Gary for review before it replaces the current screen.

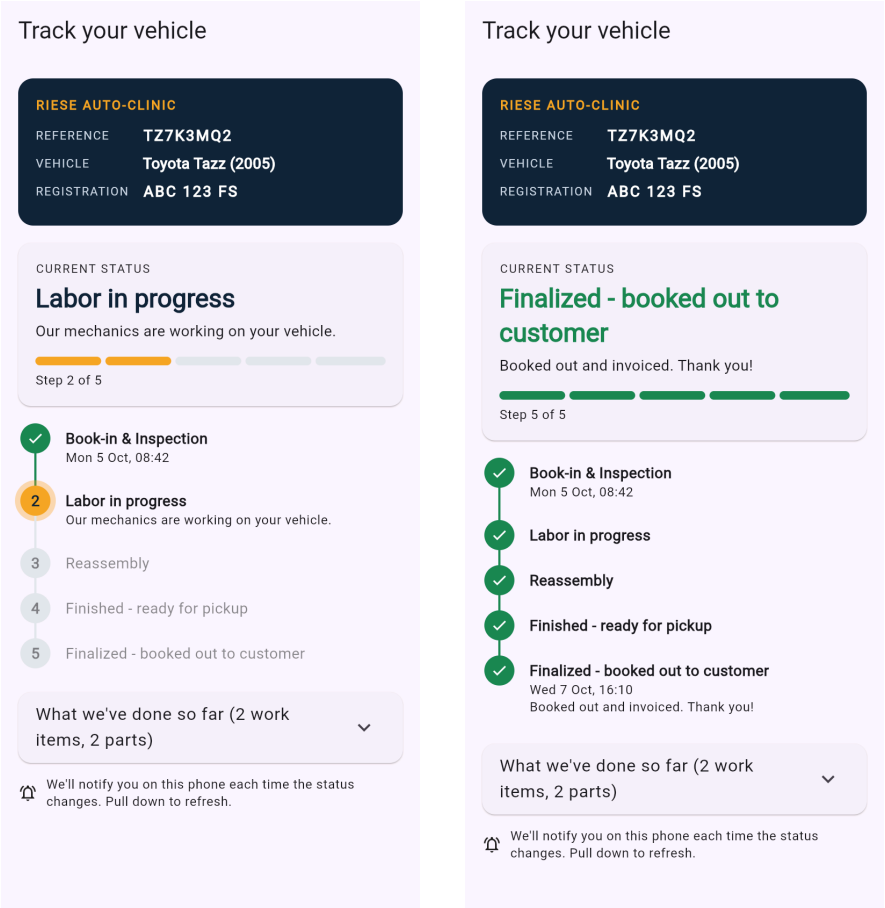


Figure 11 - Mobile customer tracking: work in progress (left) and booked out (right). Sample data; the registration is fictional.

5.6 ASP.NET Interface Design (Activity 3.3)

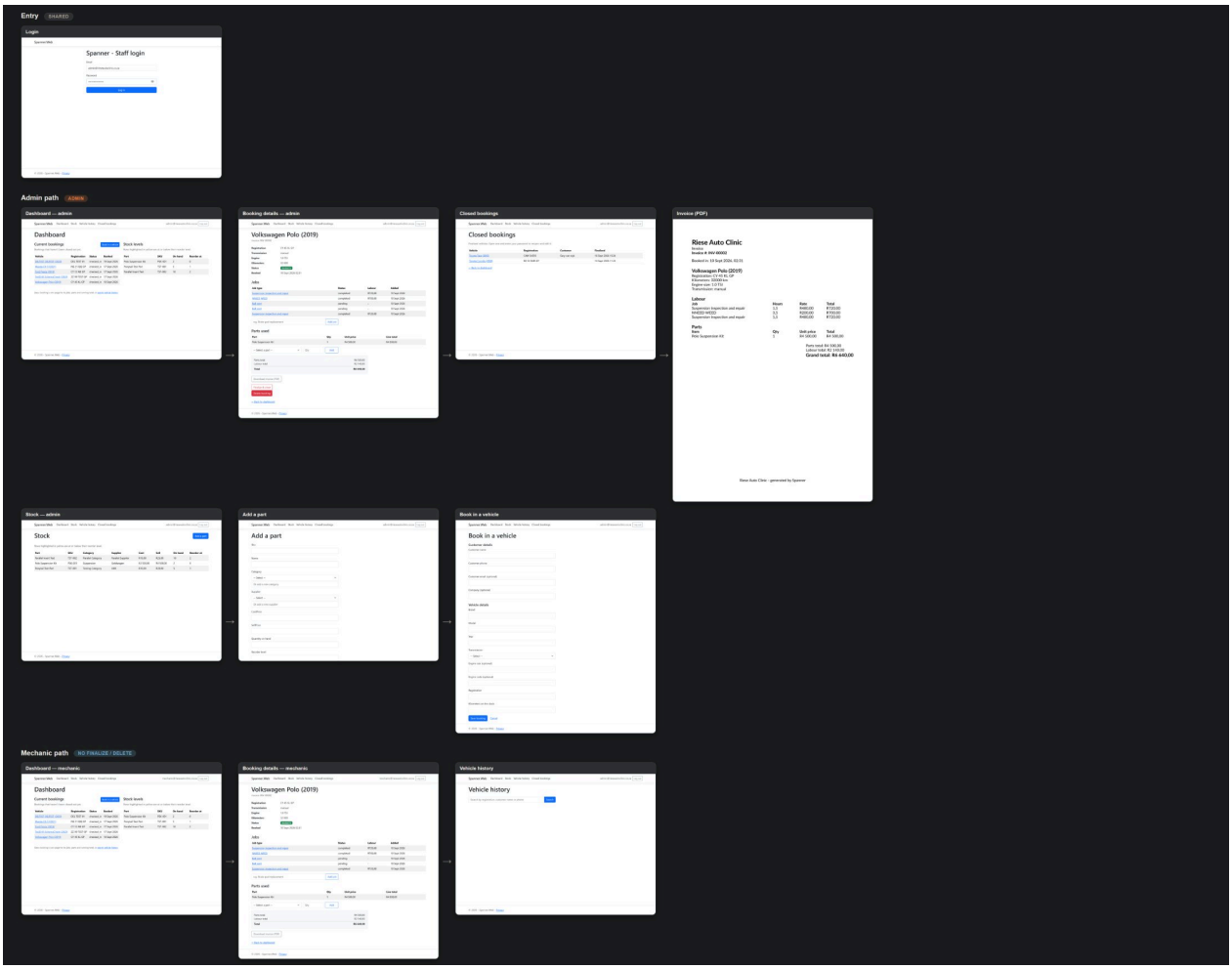


Figure 10 - ASP.NET web app screens captured from the running system, 20 Sept 2026: shared login, admin path (dashboard, booking details, closed bookings, invoice PDF, stock, add a part, book in a vehicle) and mechanic path (no finalize or delete).

Since these captures the booking page has also gained a **phase selector** (both roles, FR-15) and the admin-only Reopen.

5.6.0a ASP.NET wireframe flow by Kea (25 Sept)

This wireframe flow is Kea's own work. Kea (Keamogetse Ogone Kgosi) drew the flow of the staff web pages: from the start screen through sign-in, the dashboard, bookings, booking details, stock and the invoice, drawn as two parallel branches with arrows for the navigation (Fig. 10c). It fed into the navigation flow (Fig. 13) and the final web design.

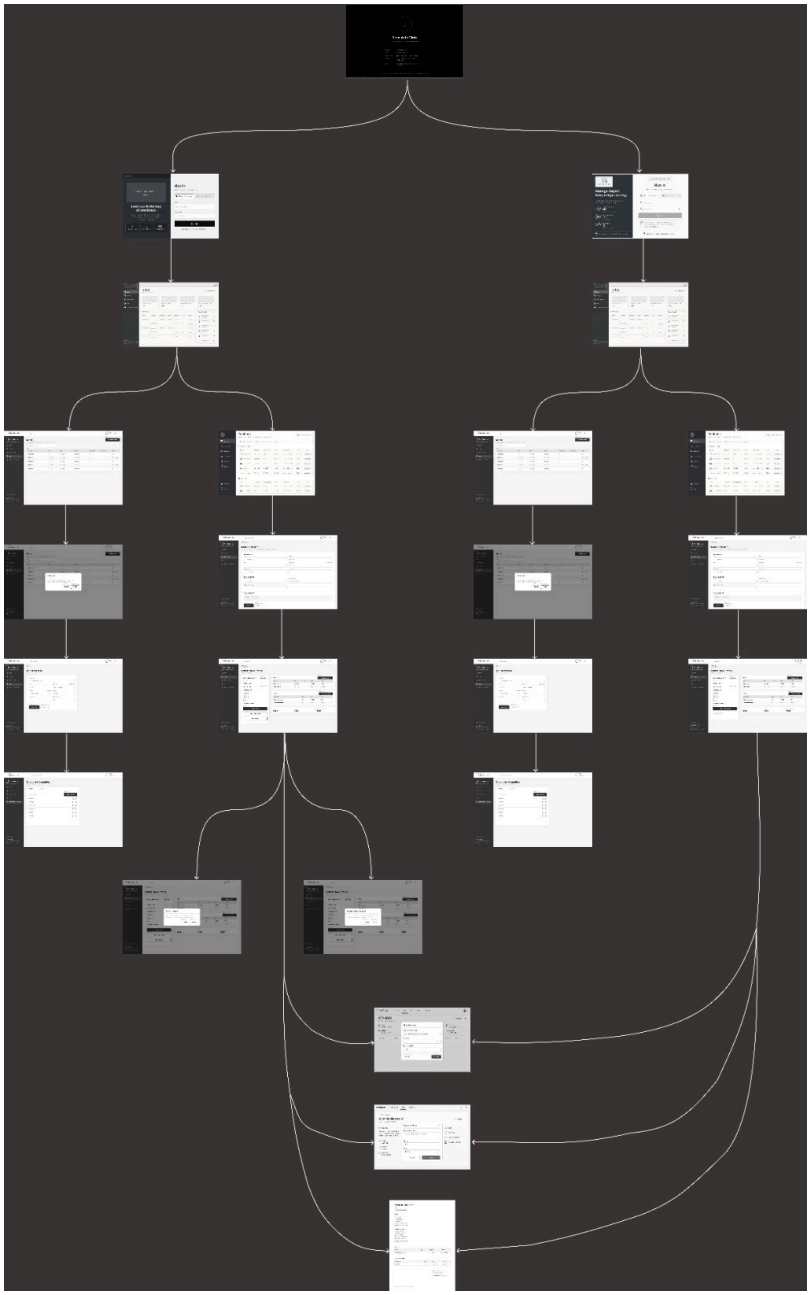


Figure 10c - ASP.NET wireframe flow by Kea (Keamogetse Ogone Kgosi), shared 25 Sept 2026. GitHub: project-docs/phase-2/designs/kea-web-wireframe.

5.6.0 Web design by Denny and the client's colours (7-13 Oct)

The web design is Denny's own work. Denny (Mathye MD) designed the web interface in **Figma**, as a clickable prototype, with an HTML prototype alongside it (Fig. 10a): a dark sidebar for navigation, a "Today" page with stat cards, a bookings table, the book-in form, stock with low-stock flags, and a phase progress bar on the booking. When we asked about colours in October, the client chose **gold and green, or gold, green and orange**. The built web app now uses Denny's layout in those colours:

Colour	Used for	Why
Dark green #173A2C	Sidebar, headers, login card	Brand colour; high contrast with white text
Gold #C9A227	Logo, active menu item, highlights, low-stock rows	Client's accent colour; draws the eye without meaning "error"
Orange #D9601A	Action buttons, the current phase	"Something to do / in progress", as in Gary's mobile mockups
Green #2E7D4F	Completed phases, "ready for pickup"	"Done", the same meaning in both apps

Two parts of Denny's prototype were not adopted, so that the design stays aligned with the approved requirements: its phase names (Inspection, Repair in progress, Quality check, Ready for collection) were replaced with the five agreed phases that the database, the mobile app and the push notifications already use (FR-15), and its SMS/WhatsApp message step was not built because customers are notified by push notification (FR-16).

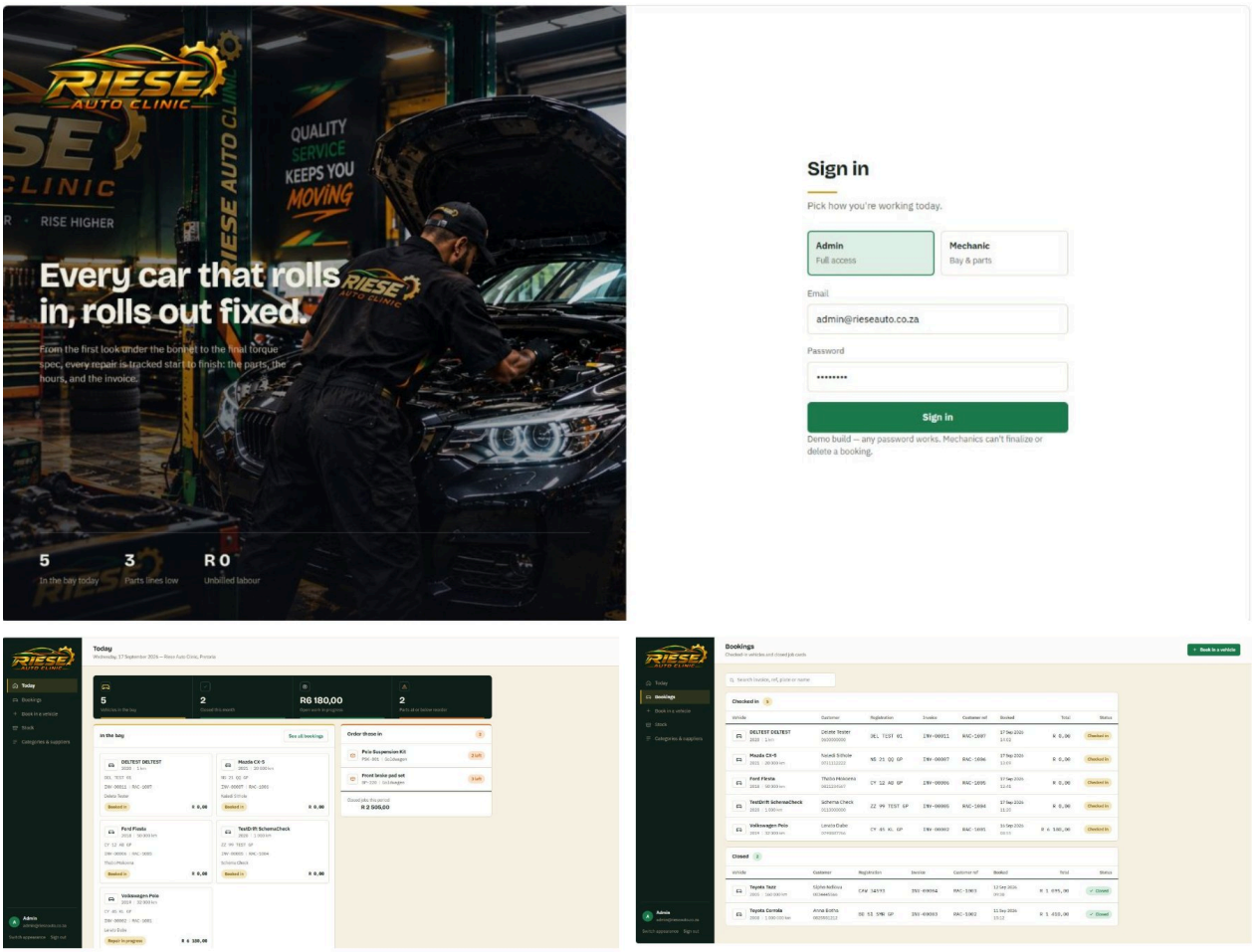


Figure 10a - ASP.NET interface designs by Denny (Mathye MD), final version in Figma, 13 Oct 2026, in the client's green and gold with the logo: A sign in, B Today, C bookings. Demo data from the prototype. Clickable prototype: https://www.figma.com/proto/yoiClYQl9oJNOD2pCRRplW/riese-auto-clinic-final-design?node-id=22-2240&t=xsFRVUIDXuvZZXIL-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1

The web pages were also redrawn in draw.io from Denny's layout and the built app (Fig. 10b), page 11 of `spanner-all-designs.drawio`.

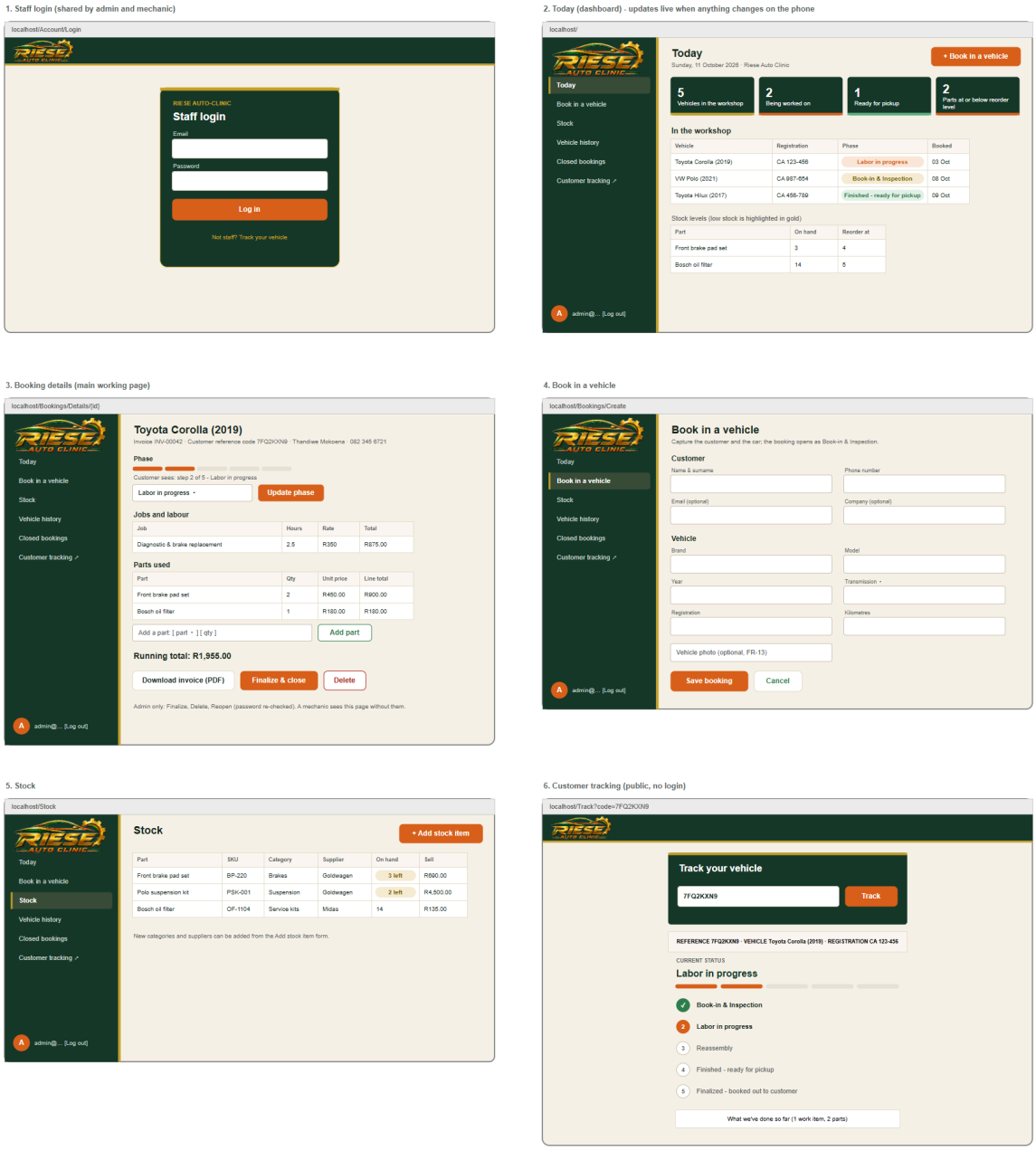


Figure 10b - ASP.NET interface design in draw.io: 1 staff login, 2 Today dashboard, 3 booking details, 4 book in a vehicle, 5 stock, 6 customer tracking (editable source: aspnet-screens.drawio).

5.6.1 Customer tracking page (web, /Track)

A public page that needs no login, linked from the staff login ("Not staff? Track your vehicle"). It uses the same layout, wording and five stages as the mobile screen and the push notification, so the customer sees one consistent story. It calls only the reference-code lookup function. It was built and tested against the live database (empty, not-found and invalid-code states); the screenshots show the found state with sample data.

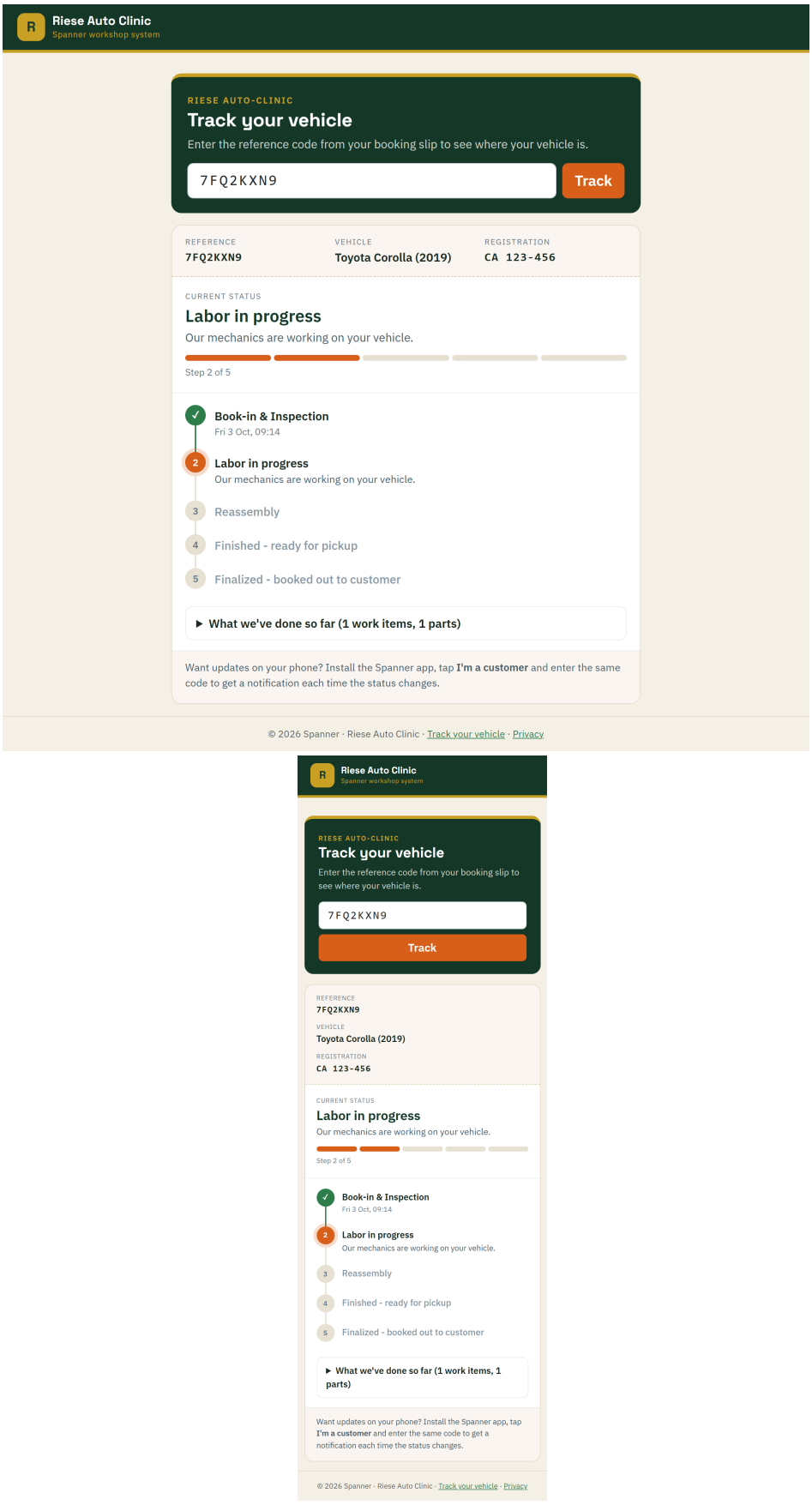


Figure 12 - Web customer tracking page in the client colours, on a laptop (left) and a phone browser (right). Captured from the running app on 11 Oct 2026 with the identifying details replaced by sample data.

5.7 Navigation Flow (Activity 3.3)

How each kind of user moves between the screens of both apps. Customers (green) never pass through a login; staff screens (blue) are the same for both roles; admin-only actions (yellow) sit on the booking details screen of each app.

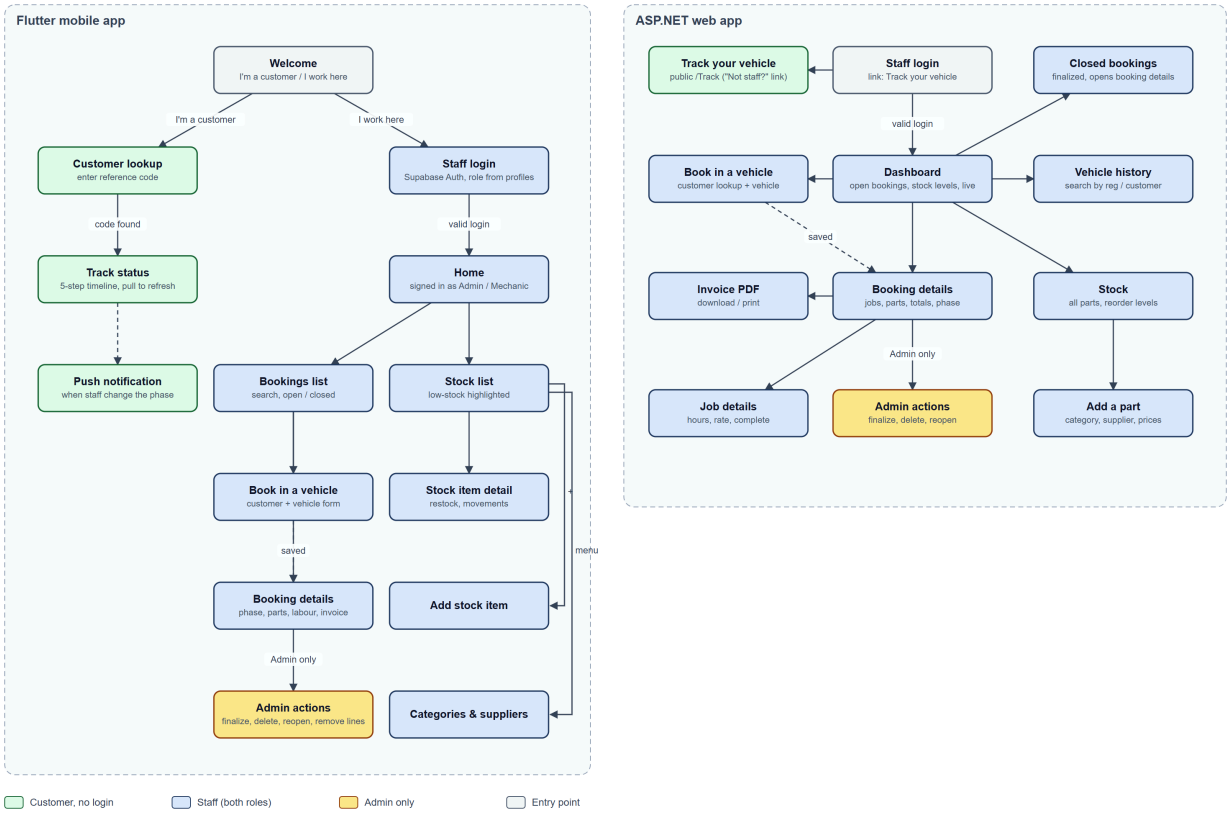


Figure 13 - Navigation flow, Flutter and ASP.NET (editable source: navigation-flow.drawio), drawn from the real routes in both code bases.

6. Design Alignment, Sources and Open Items

6.1 Design Alignment - requirement to design traceability

> Draft compiled from the design above for Denny's review (task U3-13).

Req.	Use case	Sequence	Data (ERD)	Interface
FR-01	Book in a vehicle	Fig. 3, 4	customers, bookings	Book-in screens (Fig. 9, 10)
FR-02, FR-05, FR-06	Update booking (add labour)	-	jobs / booking_labor	Booking details
FR-03, FR-10	Manage stock	-	stock_items, categories, suppliers	Stock, add a part
FR-04	Update booking (add part) incl. Decrement stock	-	booking_parts, stock_movements	Booking details
FR-07, FR-08	Download invoice	-	bookings, booking_parts, labour	Invoice PDF
FR-09	(all, via shared backend)	-	all tables	Web dashboard live refresh
FR-11	Search bookings / vehicle history	-	bookings	Vehicle history
FR-12	Login (includes Validate with Supabase Auth)	All (step 1)	profiles	Login screens
FR-13	Book in a vehicle	-	bookings.image_url	Booking details
FR-14	Check repair status (Customer)	-	bookings.reference_code	Customer tracking, mobile and web /Track (Fig. 11, 12)
FR-15, FR-16	Update booking phase	-	bookings.phase, booking_push_tokens	Phase selector on booking details (both apps)
FR-17	Admin-only use cases	Fig. 5, 6, 7	profiles	Admin vs mechanic paths (Fig. 10)
FR-18	Delete booking (includes restock)	Fig. 6	booking_parts, stock_movements	Booking details

6.2 Editable Design Sources (Appendix F)

All designs in one file: `spanner-all-designs.drawio` has one page per design (architecture, use case, five sequence diagrams, ERD, navigation flow, Flutter screens, ASP.NET screens, Gary's Flutter mockups on page 12 and Denny's web designs on page 13). It is in the ZIP and opens directly from GitHub, with no account, at:

`https://app.diagrams.net/#Uhttps%3A%2F%2Fraw.githubusercontent.com%2FBlackZ3ro%2FSpanner-Riese-Auto-Clinic%2Frefs%2Fheads%2Fphase2%2Fdesign-docs%2Fproject-docs%2Fphase-2%2Fdiagrams%2Feditable%2Fspanner-all-designs.drawio`

The brief requires a working link to, or the editable file of, every design; exported images alone do not count. draw.io (diagrams.net) is free and works online or offline, so its editable files are included in the submission ZIP under `design-sources/` and in the GitHub repository under `project-docs/phase-2/diagrams/editable/`. Open any of them at app.diagrams.net (File, Open from, Device).

Design	Tool	Editable file (ZIP and GitHub) / link
System architecture	draw.io	`architecture.drawio`

Use case diagram	draw.io; Lucidchart	`usecase.drawio` `https://lucid.app/lucidchart/fb4ebb76-2892-4f1a-888f-067c37aca98d/view`
Sequence diagrams (5)	draw.io; Lucidchart	`seq-1-book-in-vehicle-flutter.drawio, seq-1b-book-in-vehicle-web.drawio, seq-2-admin-finalize-web.drawio, seq-3-admin-delete-web.drawio, seq-4-admin-reopen-web.drawio` `https://lucid.app/lucidchart/285c1247-0f85-4e60-879c-90f48a0c91f6/view`
ERD	draw.io	`erd.drawio`
Navigation flow	draw.io	`navigation-flow.drawio`

Flutter mockups (Fig. 9a) - by Gary van Wyk	Obsidian Canvas	`Riese Auto Clinic Mockups.canvas + mockups/ (ZIP: design-sources/gary-obsidian; open the folder as an Obsidian vault)`
		`https://github.com/BlackZ3ro/Spanner-Riese-Auto-Clinic/tree/main/project-docs/phase-2/designs/gary-obsidian`
		`draw.io copy: page 12 of spanner-all-designs.drawio`

Flutter screens (Fig. 9b)	draw.io	`flutter-screens.drawio (page 10 of spanner-all-designs.drawio)`
ASP.NET screens (Fig. 10b)	draw.io	`aspnet-screens.drawio (page 11 of spanner-all-designs.drawio)`
ASP.NET wireframe flow (Fig. 10c) - by Kea	Exported image	`GitHub: project-docs/phase-2/designs/kea-web-wireframe`

ASP.NET designs (Fig. 10a) - by Denny	Figma (prototype); HTML prototype	`https://www.figma.com/proto/yoiClYQl9oJNOD2pCRRplW/riese-auto-clinic-final-design?node-id=22-2240&t=xsFRVUIDXuvZZXlL-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1`
		`GitHub: project-docs/phase-2/designs/denny-figma (HTML prototype + screenshots)`
		`draw.io copy: page 13 of spanner-all-designs.drawio`

Customer tracking screens	Flutter / ASP.NET (code)	`GitHub: web/Pages/Track; mobile branch proposal/mobile-customer-tracking`
ASP.NET screens	Running application	`GitHub: web/Pages`
Microsoft Project schedule	Microsoft Project	`TheZealotsSchedule.mpp (Khumo, 8 Oct); TheZealotsSchedule (Phase 2 update 11 Oct).xml`

6.3 Microsoft Project schedule

The schedule shows the project at the end of Phase 2. Khumo built it in Microsoft Project with the Phase 1 and Unit 3 design activities, owners, finish-to-start dependencies and milestones. It was then brought up to the actual state: every finished task at 100%, plus the development, integration and testing work (Supabase, ASP.NET, Flutter, the customer tracking page, end-to-end testing and the redesign) and the Phase 2 submission and defence (12 and 16 Oct). **Phase 3 tasks are added when Phase 3 opens**, so the schedule does not yet plan beyond 16 October. The status date is 11 October (red line). Figure 16 is the schedule in Microsoft Project; Figure 17 shows the whole schedule with % complete.

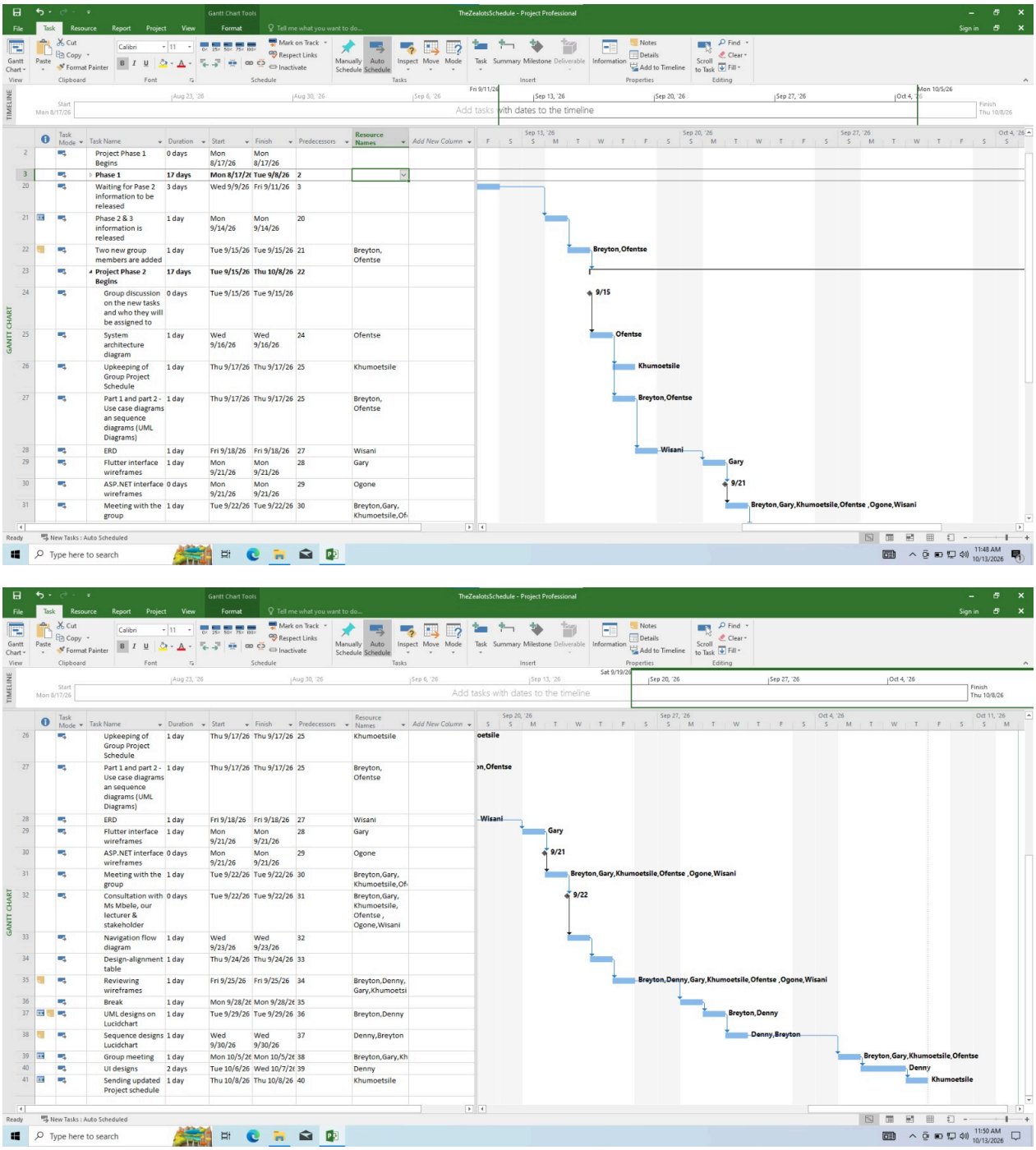
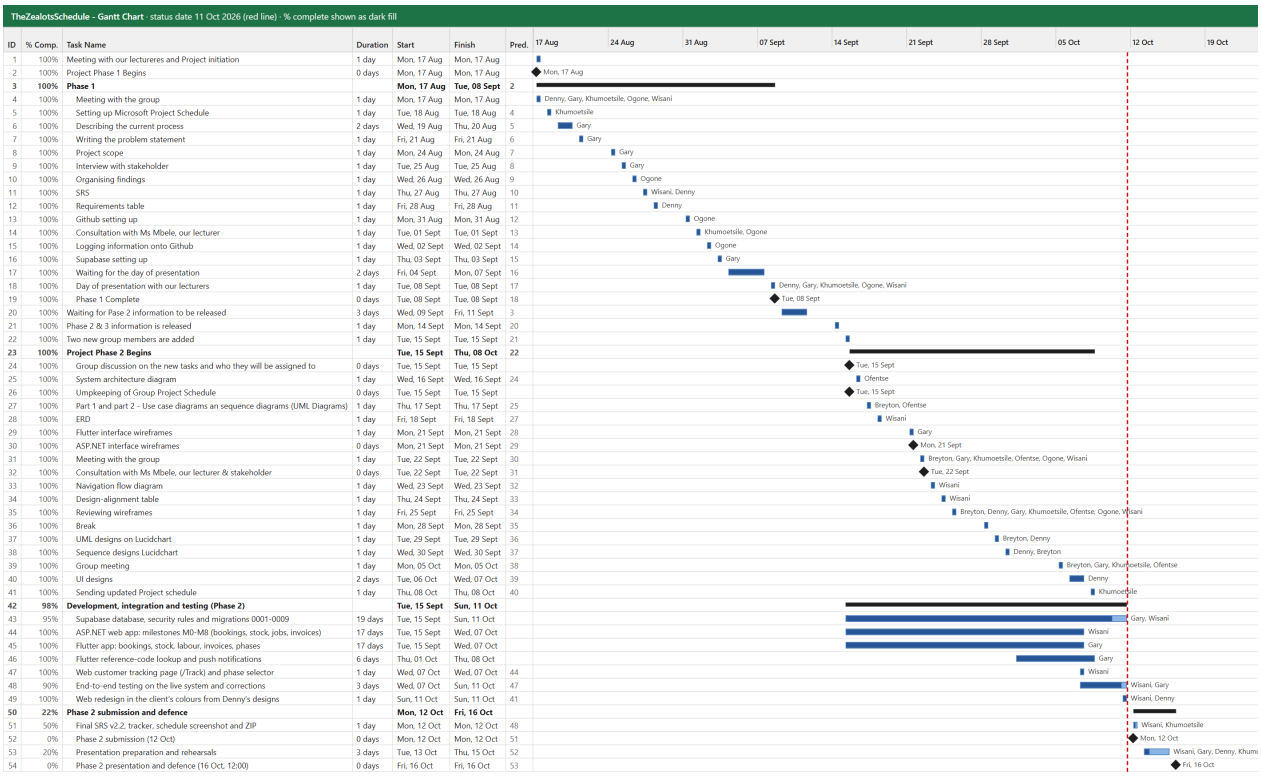


Figure 16 - Microsoft Project, Gantt Chart view of TheZealotsSchedule (screenshots by Khumo, 13 Oct 2026): Phase 2 design activities with owners, durations and finish-to-start dependencies.



ID	% Comp.	Task Name	Duration	Start	Finish	Pred.
1	100%	Meeting with our lectureres and Project initiation	1 day	Mon, 17 Aug	Mon, 17 Aug	
2	100%	Project Phase 1 Begins	0 days	Mon, 17 Aug	Mon, 17 Aug	
3	**100%**	**Phase 1**		**Mon, 17 Aug**	**Tue, 08 Sept**	2
4	100%	Meeting with the group	1 day	Mon, 17 Aug	Mon, 17 Aug	
5	100%	Setting up Microsoft Project Schedule	1 day	Tue, 18 Aug	Tue, 18 Aug	4
6	100%	Describing the current process	2 days	Wed, 19 Aug	Thu, 20 Aug	5
7	100%	Writing the problem statement	1 day	Fri, 21 Aug	Fri, 21 Aug	6
8	100%	Project scope	1 day	Mon, 24 Aug	Mon, 24 Aug	7
9	100%	Interview with stakeholder	1 day	Tue, 25 Aug	Tue, 25 Aug	8
10	100%	Organising findings	1 day	Wed, 26 Aug	Wed, 26 Aug	9
11	100%	SRS	1 day	Thu, 27 Aug	Thu, 27 Aug	10
12	100%	Requirements table	1 day	Fri, 28 Aug	Fri, 28 Aug	11
13	100%	Github setting up	1 day	Mon, 31 Aug	Mon, 31 Aug	12
14	100%	Consultation with Ms Mbele, our lecturer	1 day	Tue, 01 Sept	Tue, 01 Sept	13
15	100%	Logging information onto Github	1 day	Wed, 02 Sept	Wed, 02 Sept	14
16	100%	Supabase setting up	1 day	Thu, 03 Sept	Thu, 03 Sept	15
17	100%	Waiting for the day of presentation	2 days	Fri, 04 Sept	Mon, 07 Sept	16
18	100%	Day of presentation with our lecturers	1 day	Tue, 08 Sept	Tue, 08 Sept	17
19	100%	Phase 1 Complete	0 days	Tue, 08 Sept	Tue, 08 Sept	18
20	100%	Waiting for Pase 2 information to be released	3 days	Wed, 09 Sept	Fri, 11 Sept	3
21	100%	Phase 2 & 3 information is released	1 day	Mon, 14 Sept	Mon, 14 Sept	20
22	100%	Two new group members are added	1 day	Tue, 15 Sept	Tue, 15 Sept	21
23	**100%**	**Project Phase 2 Begins**		**Tue, 15 Sept**	**Thu, 08 Oct**	**22**
24	100%	Group discussion on the new tasks and who they will be assigned to	0 days	Tue, 15 Sept	Tue, 15 Sept	
25	100%	System architecture diagram	1 day	Wed, 16 Sept	Wed, 16 Sept	24
26	100%	Umpkeeping of Group Project Schedule	0 days	Tue, 15 Sept	Tue, 15 Sept	
27	100%	Part 1 and part 2 - Use case diagrams an sequence diagrams (UML Diagrams)	1 day	Thu, 17 Sept	Thu, 17 Sept	25
28	100%	ERD	1 day	Fri, 18 Sept	Fri, 18 Sept	27
29	100%	Flutter interface wireframes	1 day	Mon, 21 Sept	Mon, 21 Sept	28
30	100%	ASP.NET interface wireframes	0 days	Mon, 21 Sept	Mon, 21 Sept	29
31	100%	Meeting with the group	1 day	Tue, 22 Sept	Tue, 22 Sept	30
32	100%	Consultation with Ms Mbele, our lecturer & stakeholder	0 days	Tue, 22 Sept	Tue, 22 Sept	31
33	100%	Navigation flow diagram	1 day	Wed, 23 Sept	Wed, 23 Sept	32
34	100%	Design-alignment table	1 day	Thu, 24 Sept	Thu, 24 Sept	33
35	100%	Reviewing wireframes	1 day	Fri, 25 Sept	Fri, 25 Sept	34
36	100%	Break	1 day	Mon, 28 Sept	Mon, 28 Sept	35
37	100%	UML designs on Lucidchart	1 day	Tue, 29 Sept	Tue, 29 Sept	36
38	100%	Sequence designs Lucidchart	1 day	Wed, 30 Sept	Wed, 30 Sept	37
39	100%	Group meeting	1 day	Mon, 05 Oct	Mon, 05 Oct	38
40	100%	UI designs	2 days	Tue, 06 Oct	Wed, 07 Oct	39
41	100%	Sending updated Project schedule	1 day	Thu, 08 Oct	Thu, 08 Oct	40
42	**98%**	**Development, integration and testing (Phase 2)**		**Tue, 15 Sept**	**Sun, 11 Oct**	
43	95%	Supabase database, security rules and migrations 0001-0009	19 days	Tue, 15 Sept	Sun, 11 Oct	
44	100%	ASP.NET web app: milestones M0-M8 (bookings, stock, jobs, invoices)	17 days	Tue, 15 Sept	Wed, 07 Oct	
45	100%	Flutter app: bookings, stock, labour, invoices, phases	17 days	Tue, 15 Sept	Wed, 07 Oct	
46	100%	Flutter reference-code lookup and push notifications	6 days	Thu, 01 Oct	Thu, 08 Oct	
47	100%	Web customer tracking page (/Track) and phase selector	1 day	Wed, 07 Oct	Wed, 07 Oct	44
48	90%	End-to-end testing on the live system and corrections	3 days	Wed, 07 Oct	Sun, 11 Oct	47
49	100%	Web redesign in the client's colours from Denny's designs	1 day	Sun, 11 Oct	Sun, 11 Oct	41
50	**22%**	**Phase 2 submission and defence**		**Mon, 12 Oct**	**Fri, 16 Oct**	
51	50%	Final SRS v2.2, tracker, schedule screenshot and ZIP	1 day	Mon, 12 Oct	Mon, 12 Oct	48
52	0%	Phase 2 submission (12 Oct)	0 days	Mon, 12 Oct	Mon, 12 Oct	51
53	20%	Presentation preparation and rehearsals	3 days	Tue, 13 Oct	Thu, 15 Oct	52
54	0%	Phase 2 presentation and defence (16 Oct, 12:00)	0 days	Fri, 16 Oct	Fri, 16 Oct	53

Figure 17 - The whole schedule at the end of Phase 2 with % complete (dark fill), including the development, testing, submission and defence tasks. Diamonds are milestones.

6.4 Items outstanding before submission (12 October 2026)

Item	Owner
Final .mpp with % complete (the screenshot is in Figure 16)	Khumo
Run migrations 0009 then 0004 then 0008 (R-17; fixes "add part" properly); confirm the Firebase key was revoked (R-10); review the mobile tracking proposal	Gary

Part C

Implementation (Phase 3)

Added in version 3.0, 30 October 2026

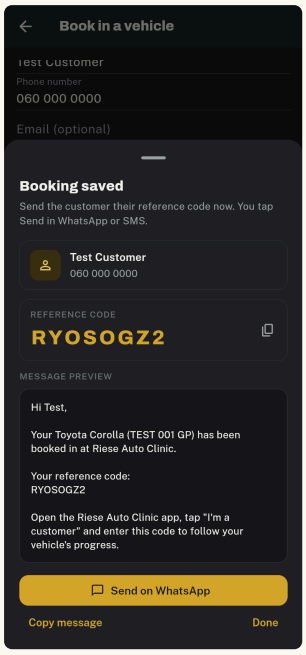
C1. What was built and the two shared processes

Spanner replaces Riese Auto-Clinic's paper and memory-based process. Mechanics and the front desk use a **Flutter mobile app** and an **ASP.NET Core 10 (Razor Pages) website**. Both talk to **one Supabase project** (PostgreSQL, Auth, Row Level Security, database functions, Realtime on the web). Customers need no account: they follow a repair with a reference code in the app or on the public `/Track` page.

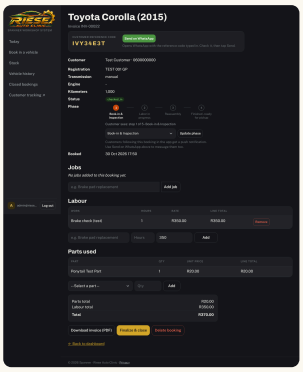
Neither app contains a service-role key. Both use the public anon key, and each staff request carries that person's own login, so database rules apply to every action. The website is hosted on Vercel; the mobile app runs on Android (and as a web build for the screenshots).

Two shared processes, shown working in both directions

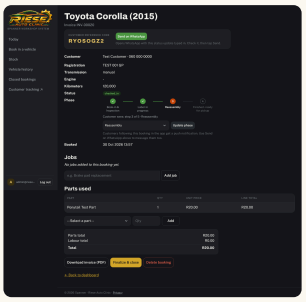
#	Process	Flutter action	ASP.NET action	Shared data
P1	Book-in, part and stock stay in step	Book in a vehicle, add a part and labor	The booking, code, customer and part are on the website; stock page shows the new quantity	`bookings`, `customers`, `booking_parts`, `stock_items`
P2	Phase change reaches the customer	Customer enters the code and sees the phase; staff list updates	Staff move the phase, press Send on WhatsApp; `/Track` shows it	`bookings.phase`, `reference_code`



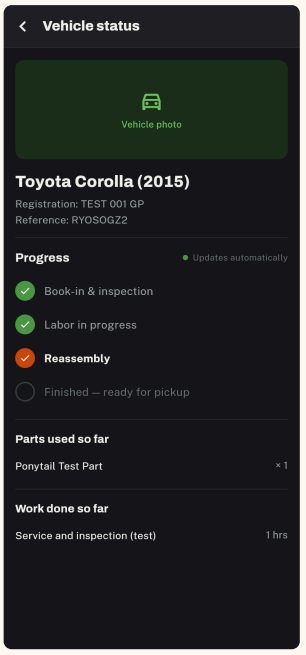
m09: booking created on the phone, reference code and WhatsApp message.



w04: the same booking, invoice number and part on the website.



w10: phase changed on the website.



m17: the phone shows the website's change.

C2. Requirement-to-function status

Status as seen working on 30 October 2026 against the live system, with dummy data only. "Partly" means it works with a stated limitation.

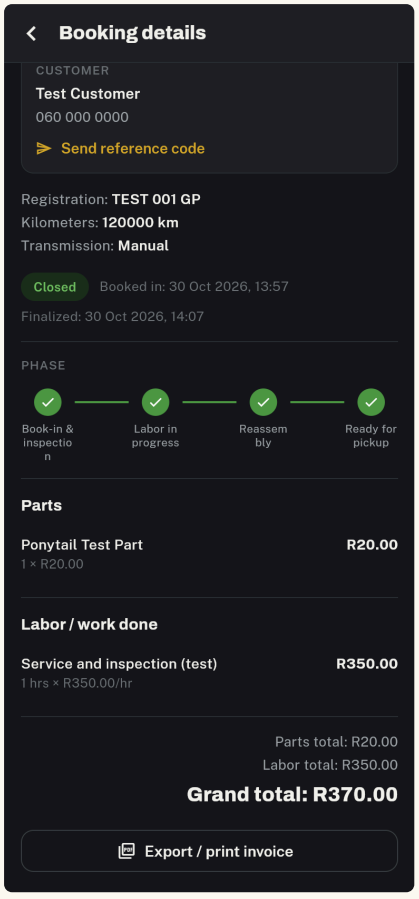
ID	Requirement	Status	Evidence	Notes
FR-01	Book in a vehicle (customer and vehicle details, validation)	**Built and shown**	Mobile m07-m10; web w05	Both apps. Reference code and invoice number are generated by the database.
FR-02	Assign jobs to a booking	**Built and shown**	Web w11	Jobs on the website; labour is no longer kept on jobs (one ledger, see section 5).
FR-03	Add and view stock items	**Built and shown**	Mobile m23; web w06	Same stock table in both apps.
FR-04	Link parts to a booking, stock drops	**Built and shown**	Mobile m11, m12, m24; web w04	Both apps now use one database function, `add_booking_part` (all-or-nothing); the phone changed on 30 Oct.
FR-05	Record labour hours and rate	**Built and shown**	Mobile m15; website Labour section	One ledger since 30 Oct: both apps write the same table, `booking_labor`.
FR-06	Job cost = parts + labour	**Built and shown**	Mobile m16; website booking page	Phone and website now read the same labour lines. Tested 30 Oct: R350 + R150 = R500 on the page and on the invoice.
FR-07	Invoice PDF	**Built and shown**	Web w16 and invoice-test-booking-30-oct.pdf; mobile export	Both apps generate a PDF from the same labour lines.
FR-08	Share / download the invoice	**Built and shown**	Web w16	Download link on the web; export / print on the phone.
FR-09	Changes seen by the other app	**Partly**	m18, m27, w04, w10	Web dashboard live-reloads. The phone has no live updates; pulling a list or booking down reloads it (m27, added 30 Oct) (L3).
FR-10	Stock levels and low-stock highlight	**Built and shown**	m23, w03, w06	
FR-11	Search and list bookings, vehicle history	**Built and shown**	m06, m22, w13	Web vehicle-history search by registration. No VIN column exists (approved change).
FR-12	Staff login and roles	**Built and shown**	m04, m25, w01, w02, w15	Supabase Auth; admin and mechanic.
FR-13	Wikimedia vehicle reference image	**Partly**	m10 (placeholder)	Works on the web; did not load in the capture environment (L5).
FR-14	Customer tracks a repair by reference code, no login	**Built and shown**	m02, m03, m17, w09, w14	Via a database function only; no table access for customers.
FR-15	Update booking phase	**Built and shown**	m13, m14, w10	Same five phases in both apps.
FR-16	Customer is notified of a phase change	**Partly**	m13; WhatsApp message file	WhatsApp click-to-chat works on both. Push needs a real Android phone and was not tested here (L4).
FR-17	Finalize, reopen, delete restocks: admin only	**Built and shown**	m16, m20, m21, m26, w12, w15	Hidden for mechanics in both apps; delete is database-enforced (0007).
FR-18	Delete a booking and restock its parts	**Built and shown**	Run on the live system; stock returned from 2 to 3	Admin only.

Non-functional requirements. NFR-01 simple interface: validation messages and a four-step tracker (m07, w02). NFR-02 speed: checked in Phase 2 testing (pages in 0.4 to 2 s, invoice PDF 1.7 s). NFR-03 behaves when Supabase is unreachable: clean message, no crash (tested earlier). NFR-06 cross-browser: only Chromium was available; Firefox and Safari were **not** tested.

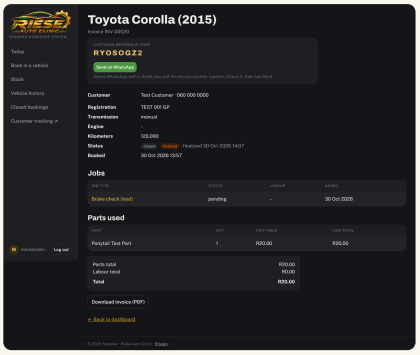
C3. Backend, access rules and roles

User	Allowed	Enforced by
Anyone not logged in	Nothing staff-only	Checked live 20 Oct: no table rows readable and the staff functions refuse (migration 0008)
Customer (no login)	Their own booking, by reference code	Security-definer function `get_booking_by_reference_code` only
Mechanic	Book in, parts, labor, phase, stock; no finalize, reopen or delete	Delete: database (0007). Finalize and reopen: database (0011). A mechanic's update to a closed booking changed 0 rows while the admin's identical update changed 1 (30 Oct)
Admin	Everything, including finalize, reopen, delete with restock	Admin check in the database

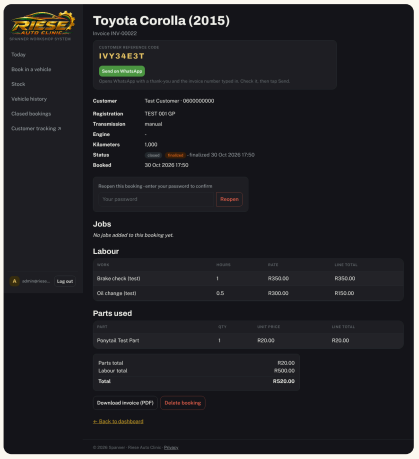
Migration status. All of 0001 to 0011 have been run on the shared project (the group confirmed 0009 and 0011 on 30 Oct; I confirmed 0008 and 0010 from outside the database).



m26: mechanic on a closed booking, no Reopen or Delete.



w15: same booking on the website as a mechanic.



w12: admin view, Reopen and Delete available.

C4. Requirements Change Log, continued (Changes 12-18)

Changes 1 to 11 are in Part A (Section 1.10) and Part B (Section 4.2). The changes below were approved while the system was being built.

No.	Date	Change	Reason and effect
12	30 Oct 2026	Customer repair-status lookup by reference code (FR-14) confirmed by the stakeholder	Michael Riese confirmed Change 7, which partly reversed Change 5. Customers see only the phase of their own booking, no login, no ETAs.
13	16 Oct 2026	Send on WhatsApp through free click-to-chat links, not the paid WhatsApp Business API	Mr G's Phase 2 feedback; keeps the zero-budget rule. Built on the website and the mobile app. Message text is checked by staff before sending.
14	Phase 2	Vehicle history searched by registration, not VIN	The live schema has no VIN column; FR-11 uses the registration.
15	30 Oct 2026	One labour ledger: `booking_labor`	The website kept labour on `jobs` and the phone on `booking_labor`, so invoices differed (R20 against R370 on one booking). The website now reads and writes `booking_labor`. Labour is a line of work, hours and rate on the booking, not on a job.
16	30 Oct 2026	Website hosted on Vercel; look matched to the mobile app	Phase 3 presentation on a real address; staff see one consistent design.
17	Oct 2026	`created_by` recorded on every booking (migration 0010)	Ms Mbele asked who booked a car in.
18	30 Oct 2026	One source for the admin role (`profiles.role`) and one-step add-part on the phone	Removes two places where the apps could disagree. Pull-to-refresh added to the phone's lists and booking page.

C5. Known limitations

#	Limitation	Effect
L1	**Resolved 30 Oct.** Labour was in two tables (`booking_labor` on the phone, `jobs` on the website).	One ledger now. Older jobs that still hold hours and rate on the `jobs` row are not counted. Screenshots w04, w12 and w16 were retaken on the live site after the change.
L2	**Resolved 30 Oct.** Admin role came from two places (mobile: `profiles.role`; web: an email list).	The website now reads `profiles.role` at login, like the phone. The email list is only a fallback when the profile cannot be read. A role change takes effect at the next login.
L3	The mobile app does not update live.	Pull a list or booking down to reload it (added 30 Oct); no automatic live updates.
L4	Push notifications and fingerprint login were not tested (need a real Android phone).	Unverified in this evidence.
L5	Vehicle reference photo did not load in the capture environment.	Placeholder shown.
L6	**Resolved 30 Oct.** Mobile add-part was three calls.	It now uses the same all-or-nothing function as the website. Tested on the phone app: part saved, stock 3 to 2, one stock movement.
L7	Container hosting on Vercel is beta; the site sleeps after about 5 minutes idle.	Open it 2 minutes before the demo. Fallback: `dotnet run` on a laptop.

L8	A Firebase service-account key was committed in September and again in October; it is still in git history.	It must be revoked in Google Cloud.

C6. Setup guidance

Website (`web/`)

Requires the .NET 10 SDK. Set the Supabase URL and the **anon key** (never the service-role key) as user secrets, then run it:

```
cd web
dotnet user-secrets set "Supabase:Url" "https://<project>.supabase.co"
dotnet user-secrets set "Supabase:AnonKey" "<anon key>"
dotnet run
```

On Vercel the same values are the settings `Supabase__Url`, `Supabase__AnonKey`, `DataProtection__Secret` (any long random text, shared by all containers) and `PublicSiteUrl`. See `project-docs/phase-3/hosting-on-vercel.md`.

Mobile app (`mobile/`)

Requires Flutter 3.x. Create `mobile/.env` with `SUPABASE_URL` and `SUPABASE_ANON_KEY` only. Then `flutter pub get` and `flutter run` (Android), or `flutter run -d chrome` for the web build. Push needs Firebase configuration on Android only.

Database

Run `supabase/migrations/0001` to `0011` in order in the Supabase SQL editor. Staff accounts are created in Supabase Auth with a matching row in `profiles`.

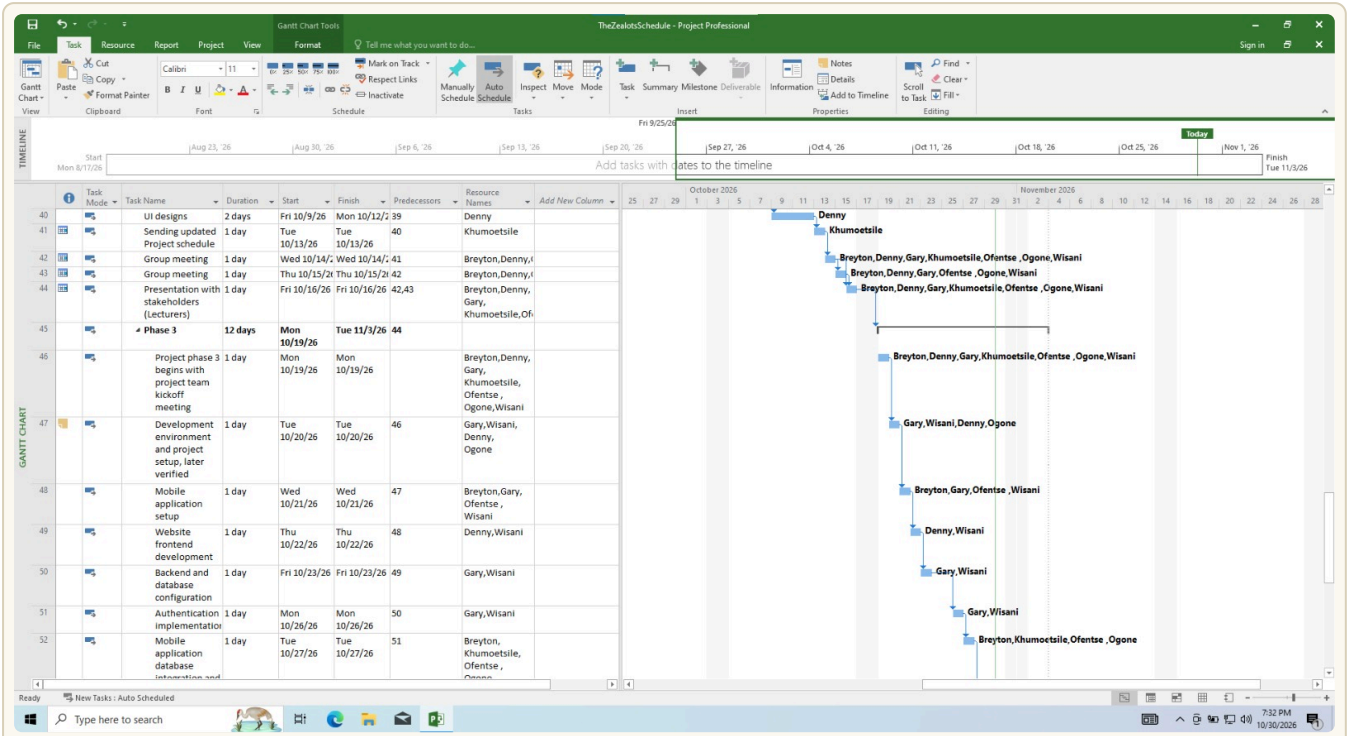
C7. Team and contributions

Member	Main contribution in Phase 3
Gary van Wyk	Flutter app, Supabase set-up and migrations, WhatsApp send sheet, biometric login
Wisani Nkuna	ASP.NET site, Vercel hosting, evidence pack, tracker, coordination
Kea Kgosi	GitHub repository and access, recording shared processes
Khumo Moremi	Microsoft Project schedule
Denny Mathye	SRS content, access rules and limitations
Ofentse Rammutla	Web function checks and screenshots
Breyton Rossouw	Flutter function checks and screenshots

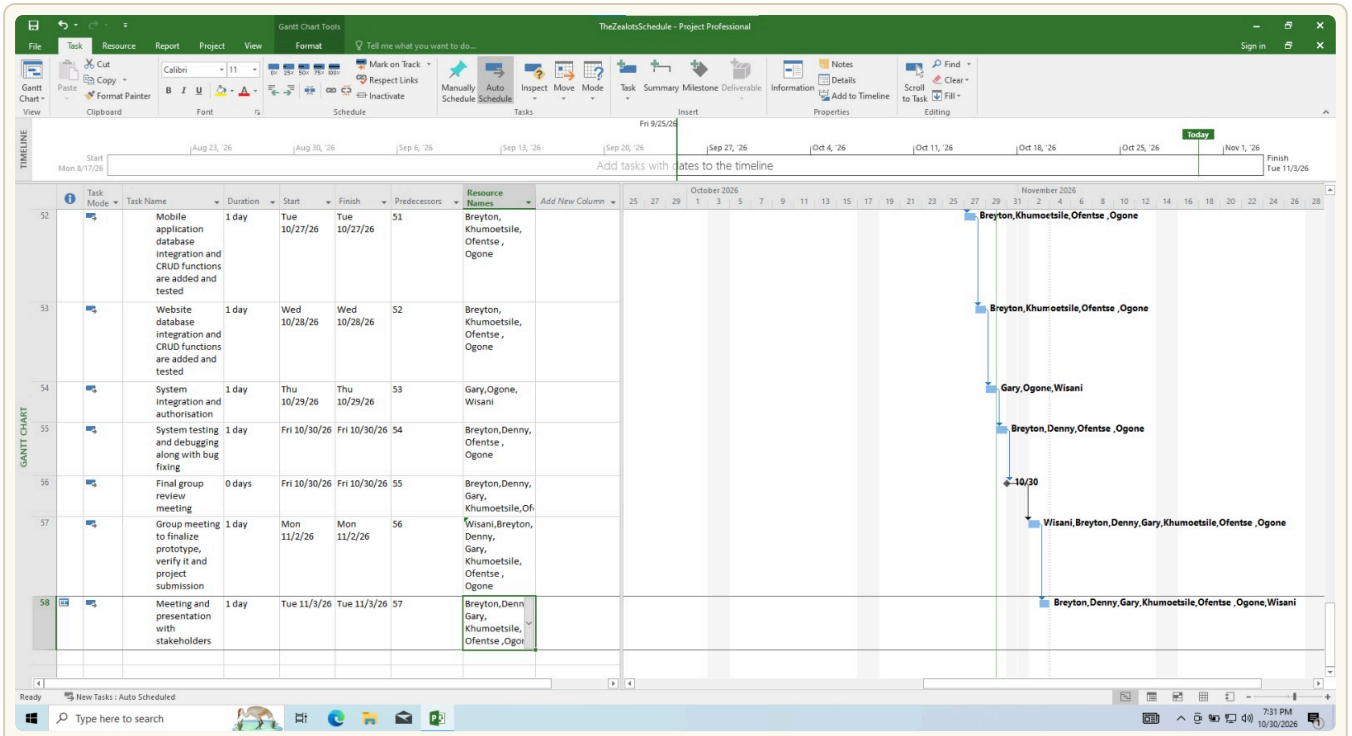
This split follows the agreed demo plan; each member confirms their own line in the tracker. Individual contribution rows are in the Phase 3 Progress Tracker; the commit history per member is on GitHub (Insights, Contributors).

C8. Microsoft Project schedule

The editable schedule `TheZealotsSchedule (Phase 3, 30 Oct 2026).mpp` is in `project-docs/phase-3/schedule/` on GitHub. Phase 3 (task 45) runs 19 Oct to 3 Nov 2026: project kickoff, environment set-up, mobile and website development, backend and authentication, database integration with CRUD, system integration and authorisation, system testing and debugging (30 Oct), the final group review meeting (30 Oct), the finalisation meeting on 2 Nov and the stakeholder presentation on 3 Nov. The screenshots show the schedule with the Today marker on 30 Oct 2026 (from the first Phase 3 file; the final file adds one meeting task, so the last two task numbers are one higher). In the saved file the Phase 3 tasks are still at 0% complete, so a copy with tasks 46 to 55 marked 100% and the status date set to 30 Oct is kept next to it as `TheZealotsSchedule (Phase 3, with progress marked, 30 Oct 2026).xml`; Khumo opens it in Microsoft Project and saves it as the final `.mpp`.



Tasks 40 to 52: end of Phase 2 and the start of Phase 3.



Tasks 52 to 58: integration, testing, final review, submission and presentation.